

Accelerating Dairy Sustainability: Integrating Economic Analysis into the Ruminant Farm Systems (RuFaS) Model for Decision Support

Documentation of Economic Data Collection and
Analytical Methods

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Overview of the RuFaS Economics Module

The Ruminant Farm Systems (RuFaS) Economics Module is designed to be a flexible and informative tool that captures key economic tradeoffs relevant to dairy farm decision-making. While the module can evaluate the economic impacts of individual or combined farm practices, it is not intended to perform a full whole-farm economic assessment that accounts for all capital investments, operating expenses, and revenue streams. Instead, it supports targeted evaluations of specific management or investment strategies by quantifying their impact on relevant costs, revenue streams, and return metrics. These strategies reflect real-world decisions made by dairy producers—from adjustments in feed rations and land use to investments in animal infrastructure and manure handling systems. Each practice or scenario is analyzed to understand its financial tradeoffs, risks, and potential benefits under varying production and market conditions, helping users explore the economic implications of farm-level change within a defined system boundary.

Relevant Farm Practices and Target Analyses

Beef-on-Dairy Breeding Strategies

This breeding strategy uses beef sires—often in combination with sex-sorted semen—to produce crossbred calves with greater economic value than conventional dairy bull calves. Crossbred beef calves typically exhibit better feed-to-gain ratios, improved carcass traits, and higher market demand, allowing them to enter premium beef markets. The practice is assessed based on the additional semen costs, the market value of crossbred calves, and the balance between maintaining adequate replacement heifers and maximizing profitability from calf sales.

High-Quality Feed for Increased Milk Production

This practice involves upgrading feed rations with more digestible forages, balanced nutrients, or energy-dense concentrates to improve milk output. The economics module evaluates the relationship between increased feed costs and gains in milk production, helping producers determine whether feed investments deliver sufficient returns under various herd productivity levels and milk pricing conditions.

Methane-Reducing Feed Strategies

Targeted adjustments to feed composition—such as the inclusion of fats, tannins, or feed additives—can lower enteric methane emissions. This practice is evaluated for its potential to generate revenue through carbon credit markets, with consideration given to both the effectiveness and cost of implementation across different feeding systems.

Feed Quality and Manure Nutrient Value

Enhancing feed quality can increase the nutrient concentration of manure, improving its agronomic value and reducing reliance on synthetic fertilizers. The economics module examines the tradeoff between feed investment and fertilizer savings by quantifying nutrient recovery from manure and its potential to meet crop nutrient needs.

Land Allocation for Feed Cost Reduction

Strategically selecting crops and allocating acreage to produce more homegrown feed—such as

corn silage or alfalfa—can reduce reliance on purchased feed and enhance on-farm feed self-sufficiency. The analysis focuses on whether shifts in land allocation strategies improve economic performance by lowering feed procurement expenses while maintaining or enhancing feed quality.

Conservation Cropping Practices

Conservation cropping encompasses a suite of field-level practices—including no-till or reduced tillage, crop rotation, and cover cropping—aimed at improving long-term soil health and farm resilience. These practices can reduce erosion, enhance water retention, and increase soil organic matter, which in turn lowers fertilizer needs and improves crop performance. Economically, these strategies may reduce fuel and labor costs associated with tillage operations, particularly under no-till systems. They can also lead to more consistent yields in the face of weather variability due to improved soil structure and moisture holding capacity.

In addition to reducing input costs, conservation cropping practices can generate revenue through participation in soil carbon credit markets. Practices that increase soil carbon stocks, such as cover cropping and reduced tillage, may be eligible for offset programs that pay producers per ton of CO₂-equivalent sequestered. The economics module evaluates the tradeoffs between implementation costs (e.g., equipment changes, seed for cover crops, additional management) and the combination of operational savings and carbon credit revenue. This practice group is particularly important for understanding how on-farm agronomic changes can align with emerging climate-smart incentives while maintaining or improving profitability.

Manure Management System Investments

Investments in manure management infrastructure—such as anaerobic digesters, solid-liquid separators, and long-term storage systems—can unlock multiple economic and environmental benefits. These systems reduce nutrient loss and enable more efficient nutrient application, lowering fertilizer costs. Anaerobic digesters can produce renewable natural gas (RNG) or electricity through combined heat and power (CHP), generating revenue from energy sales, low-carbon fuel programs, and carbon credit markets. Additional cost savings may come from reduced hauling and off-site disposal requirements. The economics module evaluates these systems by incorporating capital and operating costs, energy and co-product revenues, and nutrient and transport savings to assess their contribution to farm profitability and sustainability.

Animal Comfort Improvements

Investments in cow comfort—such as upgraded barns, enhanced ventilation, and improved bedding systems—can reduce veterinary and labor costs by lowering the incidence of health issues, reducing lameness, and improving animal handling efficiency. These improvements may also shorten milking times and reduce labor strain during routine management tasks. Enhanced comfort supports increased milk production by reducing stress-related disruptions in feeding, lying, and rumination behaviors. The economics module evaluates the cost of these infrastructure upgrades relative to their potential to improve profitability through increased milk yield and reductions in both veterinary and labor expenses.

Feed Storage Infrastructure Enhancements

Upgrading on-farm feed storage practices can play a critical role in preserving the quality and quantity of harvested or purchased feed. Poor storage conditions can result in significant dry matter and nutrient losses, leading to increased feed costs and reduced ration consistency. Improvements in storage aim to reduce shrink, spoilage, and microbial degradation, helping ensure that farm-grown feed remains usable throughout the year. This can reduce the need for additional feed purchases and contribute to greater predictability in ration formulation. The economics module evaluates these enhancements based on their potential to lower feed waste, improve feed efficiency, and reduce the overall cost of meeting herd nutritional requirements.

Flexible Economic Framework

The Flexible Economic Framework (FEF) serves as the core analytical structure of the RuFaS Economics Module. It uses a unified Discounted Cash Flow Rate of Return (DCFROR) model to evaluate the economic impacts of both individual practices and combined management strategies. Rather than relying on separate calculators for different analyses, the framework routes all analyses through a single model, adapting its assumptions and outputs based on the type of input data and user-defined scenario setup.

The FEF supports two types of economic analysis depending on whether capital investment is included in the scenario:

1. **Partial Budget Analysis (PBA)** – used to evaluate changes in operating expenses and revenues in scenarios that do not involve capital investment.
2. **Discounted Cash Flow Rate of Return Analysis (DCFROR)** – used when capital expenditures are included, allowing for evaluation of long-term cash flows, depreciation, and the time value of money.

Both analysis types are run through the same underlying model structure and use the same data sources and economic logic. The key difference lies in the financial assumptions and outputs: the PBA provides annual margin-based metrics, while the DCFROR generates investment metrics such as internal rate of return (IRR), payback period, and minimum product selling prices.

Importantly, both the PBA and DCFROR approaches focus on the financial impact of specific practices or investment strategies. They do not represent a comprehensive whole-farm economic analysis and should not be interpreted as reflecting all capital, operating, and revenue flows on the farm. Outputs are only valid within the boundaries of the modeled scenario and are intended to inform decisions at the practice or investment level.

Economic Performance Metrics

The following economic metrics are used throughout the framework to characterize the financial performance of different farm configurations:

- **Internal Rate of Return (IRR):** The discount rate at which the net present value (NPV) of all future cash flows equals zero. It reflects the expected annual rate of return on an investment and is commonly used to assess the profitability of capital-intensive projects.
- **Return on Investment (ROI):** The ratio of net benefit to cost, expressed as a percentage. ROI measures the efficiency of an investment or practice by comparing the gains (increased revenue or cost savings) to the additional expenses required to implement it.
- **Payback Period:** The amount of time (usually in years) it takes for an investment to recover its initial cost from accumulated net cash flows. Shorter payback periods generally indicate lower financial risk.
- **Profit Margin / Net Annual Cash Flow:** The difference between annual revenues and operating expenses. This metric provides a snapshot of profitability within a single year and is particularly relevant for partial budget analyses without capital investment.
- **Minimum Product Selling Price (MPSP):** The minimum price at which a product (e.g., milk) must be sold for a farm practice or investment to break even over the analysis period. This is useful in evaluating how sensitive an investment's viability is to market fluctuations.

Decision Logic and Output Routing

The model adjusts its internal logic depending on whether capital expenditures (CAPEX) are included and whether one or more scenarios are being compared:

- **Single scenario with CAPEX:** When analyzing a single farm configuration that includes capital investment, the DCFROR model incorporates the time value of money, capital depreciation, and long-term cash flow structure. Outputs from the model include net annual cash flow (profit margin), IRR, payback period, and MPSP (e.g., the lowest milk price required to break even on the specific investment or set of practices under review — not the full farm operation).
- **Single scenario without CAPEX:** If the analysis does not include capital investment, the model performs a PBA. It focuses only on changes in operating expenses and revenues, without discounting or depreciation. In this case, the model reports profit margins or net annual cash flow as the primary output.
- **Scenario comparison with CAPEX:** When comparing two scenarios that include capital investments, the model calculates the change (delta) in key financial indicators such as IRR, payback period, or MPSP. It also reports ROI between the scenarios, providing a detailed understanding of how an investment alters financial performance.
- **Scenario comparison without CAPEX:** When comparing two farm scenarios that differ in management practices but do not involve capital investments, the DCFROR model calculates the ROI between the two. This can be done for a single year, across multiple years, or as a cumulative ROI over the entire analysis period.
- **Scenario comparison with CAPEX in only one scenario:** When only one of the two scenarios includes a capital investment, full investment metrics cannot be directly compared. In this case, the model defaults to reporting the difference in net annual cash flow between the two configurations. This provides a valid, if limited, basis for comparing

the financial impact of the scenarios while avoiding misinterpretation of incompatible metrics like IRR and ROI. However, the FEF can still provide IRR, payback period, and MPSP for the scenario that does include CAPEX.

This harmonized framework allows users to conduct a wide range of economic analyses—from quick evaluations of operating tradeoffs to in-depth investment analyses—using a consistent methodology and model structure. Figure 1 illustrates the decision logic used by the FEF to determine the appropriate analysis pathway and output metrics.

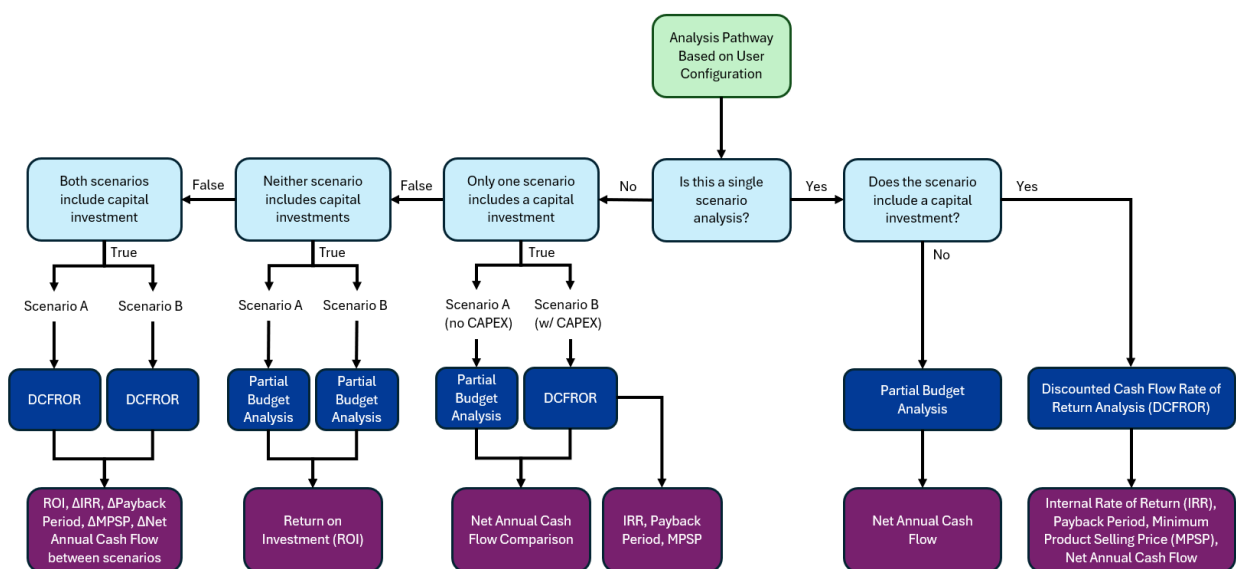


Figure 1. Logic flow of the Flexible Economic Framework (FEF), showing how user-defined scenario configurations determine analysis routing and output metrics. The decision flow distinguishes between single vs. comparative scenarios and the inclusion or exclusion of capital investment (CAPEX), guiding whether the analysis follows a partial budget or discounted cash flow approach.

Required Input Parameters

Variable Naming Convention

To ensure consistency, traceability, and flexibility across the RuFaS Economics Module, all biophysical, economic, and non-numeric input variables follow a structured naming convention using dot notation. Each variable name reflects a hierarchical structure composed of a **variable category**, an optional **subcategory**, and an **item-level variable**, as shown below:

[system-category_function].[subcategory].[item-level-variable]

Biophysical inputs are grouped into four primary system categories: **animal**, **manure_mgmt**, **crop**, and **feed_storage**. Within the first element of the naming convention, the system category is

followed by an underscore and a specified function, either **inputs**, **products**, or **variables**, depending on the role of the input parameter.

- **inputs** represent requirements to operate the system (e.g., fuel, labor, maintenance)
- **products** represent outputs of the system (e.g., milk, calves, RNG, carbon credits)
- **variables** represent intermediate values used in calculations or passed to other functions (e.g., dynamic costing equations)

Economic inputs are grouped into four system categories: **commodity_prices**, **farm_services**, **capital_costs**, and **cashflow_inputs**. These may be followed by sub-categories and item-level variables or may be directly followed by item-level variables.

The subcategory provides a more specific grouping within each system, often based on structured combinations such as [crop_form] (e.g., alfalfa_silage, alfalfa_baleage, corn_silage) or indexed variables like [comfort_investment_i] (where i = 1 to N distinct investments in animal comfort). This structure allows for disaggregation of costs, inputs, and outputs by specific components, forms, or investments within a broader system category—enabling more detailed and scalable analysis.

The item-level variable identifies the specific commodity, service, or output being measured or priced (e.g., net_fertilizer_nitrogen, diesel, milk_class_2).

In the input tables:

- **Blue text** is used for subcategories when there are multiple possible values, such as different crop forms, animal classifications, or distinct comfort investments.
- **Purple text** is used to indicate item-level variables only when there is more than one possible item within the same category or subcategory (e.g., different pricing sectors for electricity or natural gas, fertilizer components, or milk classes).
- **Black text** is used for the variable category or throughout the variable name if the variable has no subcategory or has only a single option for the item-level value. These are scalar (non-indexed) values with no additional levels of specificity.

For example:

- Example 1) **crop_inputs.alfalfa_silage.net_fertilizer_nitrogen**
 - **crop_inputs** → system category specifying required inputs for crop production
 - **alfalfa_silage** → subcategory identifying a specific crop (alfalfa) and crop form (silage) with blue text indicating there are multiple available crops and crop forms
 - **net_fertilizer_nitrogen** → item-level variable identifying the specific input (nitrogen from fertilizers) with purple text indicating there are multiple fertilizer nutrient types (nitrogen, phosphorus, or potassium)
- Example 2) **commodity_prices.natgas_industrial**
 - **commodity_prices** → system category specifying the variable is a commodity price
 - No subcategory is present; subcategory text is not used

- **natgas_industrial** → item-level variable identifying the specific commodity and sector (industrial natural gas) with purple text indicating there are multiple natural gas sectors (residential, commercial, industrial)
- Example 3) **manure_mgmt_products.digester_carbon_credits**
 - A scalar variable with no subcategory or item-level indexing; the entire variable is a single value and is shown in standard black text

This naming convention supports clarity and consistency in variable tracking, streamlines integration with the FEF and ensures transparent communication of input data structures. The color-coded formatting in the tables improves usability by helping users quickly identify levels of specificity and variable structure.

Biophysical Inputs

The economics module relies on a suite of biophysical inputs to quantify changes in operational costs and revenue streams. These inputs represent outputs from the RuFaS biophysical model – such as milk production, manure nutrient content, feed use, and greenhouse gas emissions – that are critical for evaluating the economic performance of farm practices. Most biophysical inputs are automatically generated within the RuFaS biophysical model, while others are currently provided through user input but are expected to be integrated into future versions of the RuFaS model. This flexible structure allows the economics module to evolve alongside improvements in system modeling, ensuring that key technical inputs remain accurate and up to date. Required biophysical input variables are listed in Table 1 and organized by target analyses. Unless otherwise specified, all biophysical input parameters are quantified and expressed on an annual basis.

Table 1. Required biophysical input parameters for each target analysis. Only parameters that have not been previously defined in earlier sections of the table are listed under each analysis. Variables may be shared across multiple analyses but are listed only once to avoid redundancy. The default timestep for all economic inputs is annual.

Variable	Description	Units
1. Beef on Dairy Breeding Strategies		
animal_products.[cow-class_breed_sex]	Animals sold per year categorized by class, breed, and sex (listed in Appendix A)	qty./yr
animal_inputs.[semen_type]	Semen costs per year categorized by semen type (semen_conventional, semen_sexed, or semen_beef)	straws/yr
2. High Quality Feed for Increased Milk Production		
animal_inputs.[feed_form]	Feed quantities categorized by feed ingredient and form (listed in Appendix A)	kg/yr
animal_products.[milk_class]	Total milk production categorized by milk class (listed in Appendix A)	kg/yr
animal_variables.[milk_component]	Milk components (milk_protein or milk_fat) expressed in weight percent	wt. %
3. Feed Cost vs. Enteric Carbon Credits		
animal_products.carbon_credits_enteric	Total carbon credits from enteric emissions reduction	tCO ₂ e/yr
4. Feed Cost vs. Manure Quality and Fertilizer Savings		

Variable	Description	Units
crop_inputs.[crop_form].[net_fertilizer_nutrient]	Net total fertilizer requirements for each crop type and form (listed in Appendix A) and categorized by fertilizer type (net_fertilizer_nitrogen, net_fertilizer_phosphorus, or net_fertilizer_potassium)	kg/yr
animal_products.manure_sales	Quantity of manure sold as fertilizer	kg DW/yr
animal_inputs.manure_disposal	Quantity of manure disposed (at a cost)	kg DW/yr
animal_variables.manure_percent_solids	Total suspended solids of manure that leaves the farm gate (as a product or for disposal)	%TSS
danimal_inputs.manure_disposal_transport_km	Transport distance for disposed manure	km
5. Land Allocation vs. Feed Savings		
crop_inputs.[crop_form].land_ha	Total land allocated for each crop type and form (listed in Appendix A)	ha/yr
crop_variables.land_ha_purchase	Total hectares of land that must be purchased for the current scenario	ha
crop_variables.land_ha_rented	Total hectares of land that must be rented for the current scenario	ha
crop_inputs.[crop_seeds]	Total seeds planted for each crop type (listed in Appendix A)	kg/m ² /yr
crop_inputs.[crop_form].labor_hours	Total annual labor hours required for each crop type and form (listed in Appendix A)	hours/yr
crop_inputs.[crop_form].[liquid_fuel_type]	Total annual liquid petroleum fuel consumption for each crop type and form (listed in Appendix A) and categorized by fuel type (diesel or gasoline)	L/yr
crop_inputs.[crop_form].[propane_tier]	Total annual propane consumption for each crop type and form (listed in Appendix A) and categorized by price tier (propane_residential or propane_wholesale)	L/yr
crop_inputs.[crop_form].[natgas_sector]	Total annual natural gas consumption for each crop type and form (listed in Appendix A) and categorized by natural gas sector (natgas_residential, natgas_commercial, or natgas_industrial)	kg NG/yr
crop_inputs.[crop_form].[elec_sector]	Total annual electricity consumption for each crop type and form (listed in Appendix A) and categorized by electricity sector (elec_residential, elec_commercial, or elec_industrial)	kWh/yr
crop_inputs.[crop_form].[water_type]	Total annual water consumption for each crop type and form (listed in Appendix A) and categorized by water type (water_irrigation or water_municipal)	m ³ /yr
crop_products.[crop_form]	Total quantity of each crop type and form (listed in Appendix A) sold each year	kg/yr
crop_inputs.[crop_form].[tractor_hr_size]	Annual tractor hours required for each crop type and form (listed in Appendix A) and categorized by tractor size (tractor_hr_small, tractor_hr_medium, or tractor_hr_large)	tractor hours/yr
6. Efficiency Gains from Conservation Cropping Practice		
crop_products.[farm_practice].carbon_credits_soil	Total annual soil carbon credits from each conservation practice (listed in Appendix A) relative to baseline conditions	tCO ₂ e/yr
7. Manure Management Systems		
manure_mgmt_inputs.[sub-system].labor_hours	Total annual labor hours required to operate each sub-system (separator, storage, digester, biogas_upgrading, or CHP_unit) of the manure management system	hours/yr
manure_mgmt_inputs.[sub-system].[liquid_fuel_type]	Total annual liquid petroleum fuel consumption for each sub-system (separator, storage, digester, biogas_upgrading, or CHP_unit) of the manure management system and categorized by fuel type (diesel or gasoline)	L/yr
manure_mgmt_inputs.[sub-system].[propane_tier]	Total annual propane consumption for each sub-system (separator, storage, digester, biogas_upgrading, or CHP_unit) of the manure management system and categorized by price tier (propane_residential or propane_wholesale)	L/yr

Variable	Description	Units
manure_mgmt_inputs.[sub-system].[natgas_sector]	Total annual natural gas consumption for each sub-system (separator, storage, digester, biogas_upgrading, or CHP_unit) of the manure management system and categorized by natural gas sector (natgas_residential, natgas_commercial, or natgas_industrial)	kg NG/yr
manure_mgmt_inputs.[sub-system].[elec_sector]	Total annual electricity consumption for each sub-system (separator, storage, digester, biogas_upgrading, or CHP_unit) of the manure management system and categorized by electricity sector (elec_residential, elec_commercial, or elec_industrial)	kWh/yr
manure_mgmt_inputs.[sub-system].[water_type]	Total annual water consumption for each sub-system (separator, storage, digester, biogas_upgrading, or CHP_unit) of the manure management system categorized by water type (water_irrigation or water_municipal)	m³/yr
manure_mgmt_variables.digester_population	Digester population or the total number of animal units with manure feeding into the digester	Animal Units
manure_mgmt_variables.digester_volume	Digester maximum (design) volume	m³
manure_mgmt_products.digester_rng	Annual renewable natural gas (RNG) production from the anaerobic digester and RNG upgrading system	MJ/yr
manure_mgmt_products.digester_elec	Annual electricity production from the anaerobic digester	kWh/yr
manure_mgmt_products.digester_fertilizer	Annual fertilizer production from the anaerobic digester	kg DW/yr
manure_mgmt_products.digester_bedding	Annual bedding co-product from the anaerobic digester	kg/yr
manure_mgmt_products.digester_carbon_credits	Total annual carbon credits from all digester products and activities	tCO2e/yr
8. Animal Comfort		
animal_variables.[comfort_investment_i].milk_increase	Expected increase in milk yield (as a % increase over current output) from investments in animal comfort (for i = 1 to N investments)	%
animal_inputs.[comfort_investment_i].labor_hours	Expected increase (+) or decrease (-) in total annual labor hours from investments in animal comfort (for i = 1 to N investments)	±hours/yr
animal_inputs.[comfort_investment_i].[liquid_fuel_type]	Total annual liquid petroleum fuel consumption associated with each investment in animal comfort (for i = 1 to N investments) and categorized by fuel type (diesel or gasoline)	L/yr
animal_inputs.[comfort_investment_i].[propane_tier]	Total annual propane consumption associated with each investment in animal comfort (for i = 1 to N investments) and categorized by price tier (propane_residential or propane_wholesale)	L/yr
animal_inputs.[comfort_investment_i].[natgas_sector]	Total annual natural gas consumption associated with each investment in animal comfort (for i = 1 to N investments) and categorized by natural gas sector (natgas_residential, natgas_commercial, or natgas_industrial)	kg NG/yr
animal_inputs.[comfort_investment_i].[elec_sector]	Total annual electricity consumption associated with each investment in animal comfort (for i = 1 to N investments) and categorized by electricity sector (elec_residential, elec_commercial, or elec_industrial)	kWh/yr
animal_inputs.[comfort_investment_i].[water_type]	Total annual water consumption associated with each investment in animal comfort (for i = 1 to N investments) and categorized by water type (water_irrigation or water_municipal)	m³/yr
animal_inputs.[bedding_material]	Bedding requirements per month categorized by bedding material (listed in Appendix A)	kg/month
9. Feed Storage Systems		
feed_storage_variables.[storage_investment_i].dm_loss_reduction	Expected reduction in dry matter losses (as a percentage reduction in dry matter losses over the current output) from investments in feed storage (for i = 1 to N investments)	%
feed_storage_inputs.[storage_investment_i].labor_hours	Expected increase (+) or decrease (-) in total annual labor hours from investments in feed storage (for i = 1 to N investments)	±hours/yr

Variable	Description	Units
feed_storage_inputs.[storage_investment_i].[liquid_fuel_type]	Total annual liquid petroleum fuel consumption associated with each investment in feed storage (for i = 1 to N investments) and categorized by fuel type (diesel or gasoline)	L/yr
feed_storage_inputs.[storage_investment_i].[propane_tier]	Total annual propane consumption associated with each investment in feed storage (for i = 1 to N investments) and categorized by price tier (propane_residential or propane_wholesale)	L/yr
feed_storage_inputs.[storage_investment_i].[natgas_sector]	Total annual natural gas consumption associated with each investment in feed storage (for i = 1 to N investments) and categorized by natural gas sector (natgas_residential, natgas_commercial, or natgas_industrial)	kg NG/yr
feed_storage_inputs.[storage_investment_i].[elec_sector]	Total annual electricity consumption associated with each investment in feed storage (for i = 1 to N investments) and categorized by electricity sector (elec_residential, elec_commercial, or elec_industrial)	kWh/yr
feed_storage_inputs.[storage_investment_i].[water_type]	Total annual water consumption associated with each investment in feed storage (for i = 1 to N investments) and categorized by water type (water_irrigation or water_municipal)	m³/yr

Economic Inputs

The economics module requires a set of adjustable economic inputs to quantify costs, revenues, and overall financial performance across different farm management strategies. These inputs include variables such as product prices (e.g., milk, feed, fertilizer), capital and operating costs, labor rates, and credit values (e.g., carbon offsets). To support consistent and accessible analysis, a robust set of default values is provided for all economic inputs. These defaults are based on the best available data, incorporating both temporal and geographic resolution where possible—ranging from national averages to county-level estimates, depending on data availability. While users are encouraged to modify these values to reflect site-specific conditions or local market dynamics, the default dataset allows for meaningful analysis even in the absence of detailed economic data, ensuring broader applicability and ease of use.

Table 2. Required economic input parameters organized into commodity prices, farm services, capital investments, and cash flow inputs.

Variable	Description	Units
Commodity Prices		
commodity_prices.[cow-class_breed_sex]	Animal purchase price categorized by class, breed, and sex (listed in Appendix A)	\$/kg
commodity_prices.[semen_type]	Price of a dose of semen categorized by semen type (semen_conventional, semen_sexed, or semen_beef)	\$/straw (dose)
commodity_prices.[feed_form]	Feed prices categorized by feed ingredient and form (listed in Appendix A)	\$/kg
commodity_prices.[milk_class]	Milk price received categorized by milk class (listed in Appendix A)	\$/CWT
commodity_prices.carbon_credits_enteric	Price received for carbon credits from enteric emissions reduction	\$/tCO ₂ e
commodity_prices.[net_fertilizer_nutrient]	Purchase price for fertilizer categorized by fertilizer type (net_fertilizer_nitrogen, net_fertilizer_phosphorus, or net_fertilizer_potassium)	\$/kg
commodity_prices.manure_sales	Price received for manure sold	\$/kg DW
commodity_prices.[crop_seeds]	Seed costs for each crop form (listed in Appendix A)	\$/kg

Variable	Description	Units
commodity_prices.[liquid_fuel_type]	Cost for each unit of liquid petroleum fuel consumed by fuel type (<i>diesel</i> or <i>gasoline</i>)	\$/L
commodity_prices.[propane_tier]	Cost for each unit of propane consumed and categorized by price tier (<i>propane_residential</i> or <i>propane_wholesale</i>)	\$/L
commodity_prices.[natgas_sector]	Cost for each unit of natural gas consumed and categorized by natural gas sector (<i>natgas_residential</i> , <i>natgas_commercial</i> , or <i>natgas_industrial</i>)	\$/kg NG
commodity_prices.[elec_sector]	Cost for each unit of electricity consumed and categorized by electricity sector (<i>elec_residential</i> , <i>elec_commercial</i> , or <i>elec_industrial</i>)	\$/kWh
commodity_prices.[water_type]	Cost for each unit of water consumed and categorized by water type (<i>water_irrigation</i> or <i>water_municipal</i>)	\$/m ³
commodity_prices.carbon_credits_soil	Price received for carbon credits from soil carbon sequestration	\$/tCO ₂ e
commodity_prices.digester_rng	Price received for RNG sales to the grid	\$/MJ
commodity_prices.digester_elec	Price received for electricity sales to the grid	\$/kWh
commodity_prices.digester_fertilizer	Price received for fertilizer (digestate) sold	\$/kg DW
commodity_prices.digester_bedding	Price received for bedding co-products sold	\$/kg
commodity_prices.digester_carbon_credits	Price received from carbon credits from digester activities (renewable energy credits, fertilizer offsets, etc.)	\$/tCO ₂ e
commodity_prices.[bedding_material]	Cost of bedding categorized by bedding materials (<i>listed in Appendix A</i>)	\$/kg
commodity_prices.rng_upgrading_consumables	Cost of consumables (excluding fuels, electricity, and water) required for biogas upgrading to produce RNG	\$/MJ RNG
<i>Farm Services and Maintenance</i>		
farm_services.vet_costs.[cow-class_breed_sex]	Veterinary costs categorized by class, breed, and sex (<i>listed in Appendix A</i>)	\$/ (animal*yr)
farm_services.manure_disposal	Cost of manure disposal (transport and disposal)	\$/kg DW
farm_services.land_ha_rent	Cost to rent farmland per hectare per year	\$/ (ha*yr)
farm_services.labor_hours	Hourly cost for hired labor on the farm	\$/hour
farm_services.[tractor_size]	Total operating cost per tractor hour categorized by tractor size (<i>tractor_small</i> , <i>tractor_medium</i> , or <i>tractor_large</i>)	\$/tractor hour
manure_mgmt_inputs.[sub-system].annual_maintenance	Annual maintenance costs for each sub-system (<i>separator</i> , <i>storage</i> , <i>digester</i> , <i>biogas_upgrading</i> , or <i>CHP_unit</i>) of the manure management system	\$/yr
animal_inputs.[comfort_investment_i].annual_maintenance	Annual maintenance costs associated with each investment in animal comfort (<i>for i = 1 to N investments</i>)	\$/yr
feed_storage_inputs.[storage_investment_i].annual_maintenance	Annual maintenance costs associated with each investment in feed storage (<i>for i = 1 to N investments</i>)	\$/yr
<i>Capital Investments</i>		
capital_costs.land_ha	Purchase price of farmland per hectare	\$/ha
capital_costs.[sub-system]	Capital cost of each sub-system (<i>separator</i> , <i>storage</i> , <i>digester</i> , <i>biogas_upgrading</i> , or <i>CHP_unit</i>) of the manure management system. If users do not opt to use the default values (calculated with the dynamic sizing functions) or provide their own values, capital costs default to \$0 (assumes equipment is already owned)	\$
capital_cost.[comfort_investment_i]	Total capital cost of investments in animal comfort (<i>for i = 1 to N investments</i>)	\$
capital_cost.[storage_investment_i]	Total capital cost of investments in feed storage (<i>for i = 1 to N investments</i>)	\$
<i>Cash Flow Inputs</i>		

Variable	Description	Units
cashflow_inputs.target_irr	Target internal rate of return (IRR)	%
cashflow_inputs.target_payback_period	Target payback period for the investment being evaluated	years
cashflow_inputs.target_product.[item_level_variable]	The <i>item_level_variable</i> for which the minimum product selling price is determined	[<i>item_level_variable</i>]
cashflow_inputs.target_MBSP	Target minimum product selling price (MBSP)	\$/unit product
cashflow_inputs.[depreciation_i]	Annual depreciation (for <i>i = 1 to N years</i>). The cash flow defaults to the appropriate depreciation schedule for dairy farm financial analysis, however users may override the default values.	% per year
cashflow_inputs.tax_rate	Annual income tax rate	%
cashflow_inputs.loan_term	Loan term in years	years
cashflow_inputs.loan_int	Loan interest rate	%
cashflow_inputs.const_term	Construction term in years	years
cashflow_inputs.const_int	Construction loan interest rate	%
cashflow_inputs.[const_rate_i]	Construction rate (% of construction completed in each year of the construction term, for <i>i = 1 to N years</i>)	%
cashflow_inputs.equity	Total project equity (% of total capital expenses)	%

Binary Inputs and Booleans

The economics module includes a small set of targeted binary and Boolean inputs specifically related to anaerobic digester system configuration and costing methods. These user-defined selections govern how the model interprets digester design choices—such as construction materials and installation approach—and whether to apply default operational costing functions from the literature. While limited in number, these logic flags are critical for aligning the model's capital and operational cost calculations with the actual design assumptions of the scenario being evaluated. Users should adjust these inputs to accurately reflect their system configuration and desired costing approach.

Table 3. User-defined logic inputs and Booleans for digester system configuration and costing.

Variable	Description	Format
manure_mgmt_variables.digester_farm_type	Anaerobic digester farm type designation (1 = Dairy, 0 = Other)	Binary
manure_mgmt_variables.digester_is_below_ground	Anaerobic digester is below ground (TRUE / FALSE)	Boolean
manure_mgmt_variables.digester_is_concrete	Anaerobic digester is primarily constructed from concrete (TRUE / FALSE)	Boolean
manure_mgmt_variables.digester_is_steel	Anaerobic digester is primarily constructed from steel (TRUE / FALSE)	Boolean
manure_mgmt_variables.digester_use_AVC_fcn	Override user inputs for digester labor, fuels, electricity, and materials and use the variable operational costing function from Cowley and Brorsen (TRUE / FALSE)	Boolean

The variables listed in Table 3 represent the only binary and Boolean inputs currently required by the economics module. As the model evolves, additional logic inputs may be incorporated to accommodate expanded system configurations and emerging practices.

Economics Module Inputs Manager

The Economics Module Inputs Manager is responsible for organizing, interpreting, and formatting all input data used within the economic analysis in RuFaS. Its primary function is to standardize data from multiple sources—user inputs, RuFaS module outputs, and default economic values—into a consistent format that can be used in partial budget and discounted cash flow analyses. All economic variables are classified based on their role in the simulation, following a naming convention that designates either costs (e.g., *_inputs) or revenues (e.g., *_products). For example, animal_inputs refers to costs associated with animal production, while animal_products includes revenue-generating outputs such as milk or meat.

This classification allows the module to intelligently determine whether a given variable should be treated as a cost or a revenue source. In partial budget scenarios, this logic is extended to handle changes between baseline and alternative scenarios: reductions in revenue (e.g., lower milk output) are treated as additional costs, while reductions in cost (e.g., feed savings) are treated as effective revenues. This flexible interpretation ensures consistency in how net changes are evaluated across different scenarios and modules.

Each input to the economics module must have two components: a quantity (typically generated from the RuFaS biophysical module or specified by the user) and an associated price or cost (sourced from the RuFaS Economic Database). The Economic Inputs Manager multiplies these to calculate net costs or revenues. RuFaS-generated inputs offer the highest degree of accuracy within the current system configuration. However, if a RuFaS output is unavailable, the module can fall back to user inputs or predefined default values to ensure the analysis proceeds without interruption.

Some variables require internal conversions or pass-through calculations before they can be used in economic assessments. For instance, milk pricing may depend on fat- or protein-adjusted milk equivalents. In these cases, the Inputs Manager uses milk composition variables (e.g., percent milk fat or protein) to calculate a normalized product volume, ensuring consistent revenue estimates across scenarios with different milk quality outputs.

Historical Data, Inflation Adjustment, and Discounting

When historical economic data is used—particularly in DCFROR analyses—the Inputs Manager adjusts for inflation by indexing all values to a common dollar year. All historical values in the economic database are reported in nominal terms for the year they were observed (e.g., 2020 dollars for data from 2020). To ensure compatibility with DCFROR logic, the Inputs Manager deflates or inflates these values to match the starting year of the simulation. For example, if the simulation spans 2020–2024, all economic inputs will be adjusted to 2020 dollars before applying

discounting using Consumer Price Index data from the United States Bureau of Labor and Statistics [1,2].

Users conducting a price volatility analysis can also provide a vector of input prices across years. In this case, the Inputs Manager assumes the user has already adjusted prices to the appropriate dollar year and will require a flag indicating whether all values are in the same currency year. Failure to standardize dollar years across inputs may result in inconsistent or misleading DCFROR results, so users must ensure consistency before submitting vectors of price data to evaluate market volatility.

Dynamic Costing Functions

To enable robust techno-economic analysis of advanced manure management systems, this section presents dynamic capital cost functions that estimate installed costs of individual technologies or components across a range of system scales. The approach provides a structured, modular framework for estimating capital expenditures for key sub-systems—beginning with primary solids separation, progressing through anaerobic digestion, biogas utilization (via combined heat and power or RNG upgrading), and concluding with final digestate separation.

Manure Storage

Manure storage system costs are not included as default inputs in the model due to the substantial variability in design, scale, and site-specific requirements across farms. Because of this, cost data for storage system upgrades were not collected for this analysis. Instead, the model allows users to manually input capital and operational costs for their specific storage configurations if they wish to evaluate potential economic impacts of storage changes. In contrast, the model provides built-in, cost-scaling functions for manure management system upgrades that directly improve operational efficiency and value generation. These components are fully parameterized, enabling consistent, scale-sensitive techno-economic evaluation across a range of system configurations.

Anaerobic Digesters

Digester Capital Costing Function

The capital cost of the anaerobic digester is derived from a fixed cost function reported by Cowley and Brorsen [3], which provides a generalizable method for estimating digester capital costs based on system energy output. The function is expressed on a per-unit-energy basis and is formulated using survey data from 45 operational digesters across the United States, capturing both economies of scale and real-world variation across farm sizes and configurations [3]. Importantly, the function is representative of both complete mix and plug flow digester systems, making it broadly applicable to a range of farm designs. This makes it especially well-suited for national-level or scenario-based modeling, where site-specific data is not readily available. The costing function from Cowley and Brorsen [3] contains the following inputs and parameters:

Table 4. Inputs required for the anaerobic digester capital costing function from Cowley and Brorsen [3].

Input or Variable Name	Symbol	Default Value (If Applicable)	Units
<i>Function Inputs (from Economic Inputs Manager)</i>			
manure_mgmt_variables.digester_population	A_j	NA ¹	AU
manure_mgmt_variables.digester_volume	V_j	NA ¹	m ³
manure_mgmt_variables.digester_farm_type	F_j	1	Binary
manure_mgmt_variables.digester_is_below_ground	G_j	1	Binary
manure_mgmt_variables.digester_is_concrete	C_j	1	Binary
manure_mgmt_variables.digester_is_steel	S_j	0	Binary
<i>Function Constants</i>			
Intercept Term	α	10.4	\$/MJ
Construction Cost per AU	R_1	-0.413	\$/AU
Construction Cost per Volume	R_2	-0.147	\$/m ³
Farm Type Constant	ψ_1	-1.085	unitless
Below Ground Constant	ψ_2	-0.582	unitless
Concrete Constant	ψ_3	0.519	unitless
Steel Constant	ψ_4	-0.007	unitless
Standard Error Term	ε_j	0.35	\$/MJ

¹The Economics Module does not offer default values for digester population and volume. These parameters should be tied to the farm size with values provided by the RuFaS biophysical module. However, the user can still override these and supply custom or arbitrary values (for scenario analysis, sensitivity analysis, etc.).

With the inputs defined in Table 4, the annual fixed capital cost of the digester (in \$ per MJ per year) can be calculated by Equation 1:

$$AFC_{ij} = EXP(\alpha + (R_1 \times \ln(A_j)) + (R_2 \times \ln(V_j)) + (\psi_1 \times F_j) + (\psi_2 \times G_j) + (\psi_3 \times C_j) + (\psi_4 \times S_j) + \varepsilon_j) \quad (1)$$

To determine total installed capital cost as a single value (rather than an annualized metric), this value is unannualized by dividing by the capital recovery factor (CRF). Assuming a 5% discount rate and 20-year project life, consistent with the original study assumptions, the CRF is calculated using Equation 2:

$$CRF = \frac{r(1+r)^t}{(1+r)^t - 1} \quad (2)$$

where r is the discount rate (5%) and t is the project lifetime (20 years). The total installed capital cost of the digester is then determined by Equation 3 and passed on to the FEF for use in the DCFROR analysis.

$$CAPEX_{digester} = \frac{AFC_{ij}}{CRF} \quad (3)$$

Capital costs for anaerobic digesters across a range of herd sizes are shown in Figure 2. These estimates are based on an assumed manure excretion rate of 35 gallons per cow per day, a manure density of 8.4 pounds per gallon, and a hydraulic retention time (HRT) of 17 days [3,4].

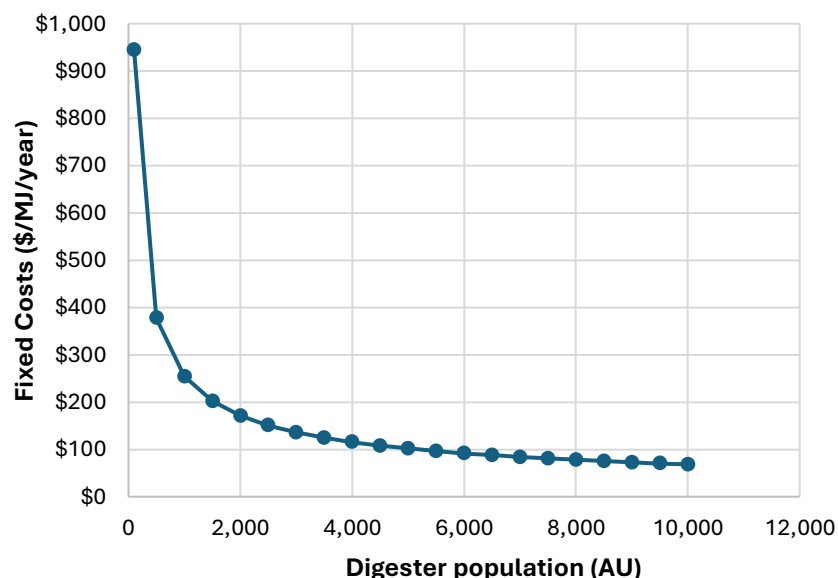


Figure 2. Fixed capital cost function for anaerobic digesters as a function of animal unit population, showing economies of scale and the characteristic cost inflection point ("knee of the curve") around 1,500 animal units [3].

This curve clearly demonstrates economies of scale, with notable reductions in unit capital cost as system size increases, particularly up to approximately 1,000 animal units—a trend that aligns well with findings from other published studies on digester economics [3,4].

Digester Operational Costing Function

Operational expenses (OPEX) in the model are quantified as the sum-product of all services and commodities required to run each subsystem of the manure management system, including material inputs, water, electricity, fuels, labor, and maintenance. For anaerobic digesters specifically, users have the option to enable a dedicated variable OPEX costing function derived from Cowley and Brorsen [3]. This is controlled through a Boolean parameter in the model `manure_mgmt_variables.digester_use_AVC_fcn`), allowing flexibility to either use user-defined inputs or the literature-based function. When activated, the variable OPEX function requires the following inputs:

Table 5. Inputs required for the anaerobic digester capital costing function from Cowley and Brorsen [3].

Parameter Name	Symbol	Default Value (If Applicable)	Units
<i>Function Inputs (from Economic Inputs Manager)</i>			
manure_mgmt_variables.digester_use_AVC_fcn	T	0 ¹	Binary
manure_mgmt_variables.digester_population	A_j	NA ²	AU
manure_mgmt_variables.digester_farm_type	F_j	1	Binary
manure_mgmt_variables.digester_is_below_ground	G_j	1	Binary
manure_mgmt_variables.digester_is_concrete	C_j	1	Binary
manure_mgmt_variables.digester_is_steel	S_j	0	Binary
<i>Function Constants</i>			
Intercept Term	β	3.531	\$/MJ/yr
Operating Cost per AU	ω_1	-0.321	\$/AU
Farm Type Constant	ϕ_1	-0.686	unitless
Below Ground Constant	ϕ_2	-0.418	unitless
Concrete Constant	ϕ_3	0.488	unitless
Steel Constant	ϕ_4	-0.113	unitless
Standard Error Term	ϵ_j	0.237	\$/MJ/yr
¹ The default setting is to not use the AVC function (0 or FALSE) and rely on biophysical inputs and user data to determine total OPEX. ² The Economics Module does not offer default values for digester population and volume. These parameters should be tied to the farm size with values provided by the RuFaS biophysical module. However, the user can still override these and supply custom or arbitrary values (for scenario analysis, sensitivity analysis, etc.).			

The annual variable operational cost (in \$/MJ/year) is then calculated using Equation 4 and passed on to the FEF for use in the DCFROR analysis.

$$AVC_{ij} = T \times EXP(\beta + (\omega_1 \times \ln(A_j))) + (\phi_1 \times F_j) + (\phi_2 \times G_j) + (\phi_3 \times C_j) + (\phi_4 \times S_j) + \epsilon_j \quad (4)$$

Figure 3 presents the modeled variable operational costs across digester populations, based on the same manure excretion rate, manure density, and hydraulic retention time used to generate Figure 2.

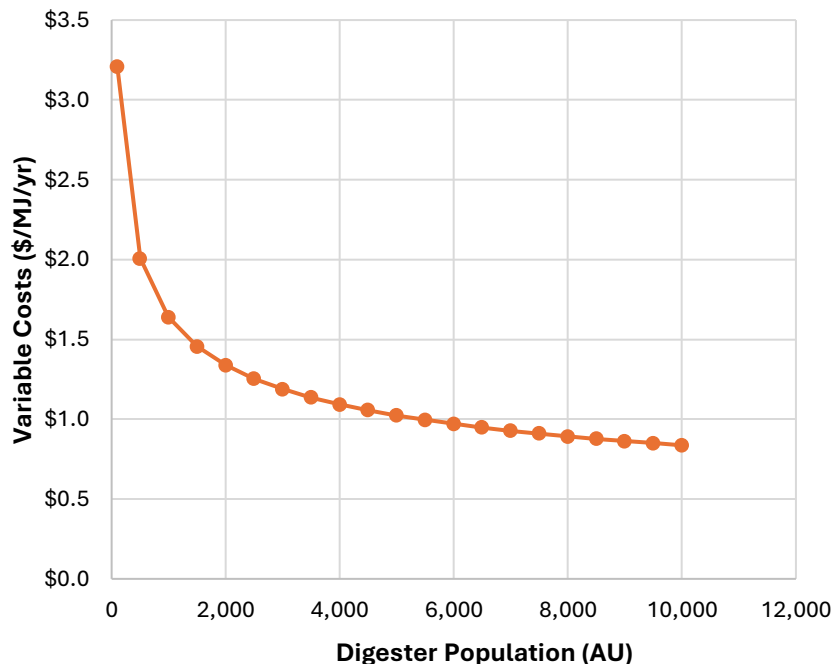


Figure 3. Variable operational expenses for anaerobic digesters as a function of animal unit population, illustrating the decreasing cost trend with increasing system size due to economies of scale and operational efficiencies [3].

Additional Manure Management System Upgrades

The manure management system is composed of modular technology components that work together to improve operational efficiency, reduce environmental impact, and capture additional value from manure streams. Each component plays a distinct role within the integrated system, from initial solids separation to energy recovery and final digestate processing, described below:

- **Primary Solids Separators (Screw Press or Rotary Screen):** Mechanical systems used at the front end of the manure management process to separate solids from liquid manure. Screw press separators use a rotating screw inside a screen for compression and dewatering, while rotary screen separators use a rotating drum for initial coarse separation.
- **Biogas Utilization (CHP or RNG Upgrading):** Biogas produced by the digester can be combusted for electricity and heat in a combined heat and power (CHP) unit or upgraded to renewable natural gas (RNG) through removal of impurities and compression to allow pipeline injection.
- **Final Digestate Solids Separator:** Following digestion, a second-stage separator (using the same costing functions as above) can be used to recover solids from the digestate stream for land application or to produce a bedding co-product.

These components of the manure management system are costed using a standardized equipment costing function that applies a scaling relationship between system size and capital

cost, along with an installation factor to account for site-specific deployment considerations. The installed capital cost for each manure management system upgrade is calculated by Equation 5:

$$C_{install} = C_{p,base} \times \left(\frac{V_m}{V_{m,base}} \right)^n \times F_{install} \quad (5)$$

Where C_p represents the purchase capital cost of the screw press separator in USD, $C_{p,base}$ is the base purchase cost of the equipment in USD at the reference flowrate, V_m is the reference flowrate, which changes based on the sub-system being costed. $V_{m,base}$ is the base flowrate, and n is the scaling exponent which also changes based on sub system being evaluated. The installation factor, $F_{install}$, accounts for labor, electrical, foundations, piping, and controls required for equipment installation. Baseline flowrates, purchase costs, scaling exponents, and installation multipliers for each manure management system upgrade are provided in Table 6 below.

Table 6. Capital cost parameters for manure management system upgrades, including baseline flowrates, base purchase costs, scaling exponents, and installation multipliers for each sub-system. These parameters are used in the standardized equipment costing function (Equation 5) to dynamically estimate installed capital costs based on system throughput.

Equipment	Base Flowrate ($V_{m,base}$)	Base Purchase Cost ($C_{p,base}$)	Scaling Exponent (n)	Install Multiplier ($F_{install}$)	Reference
Screw Press Separator	100 gal/min	\$75,000	0.65	1.4	USDA NRCS (2018), EPA AgSTAR (2021), Vendor Data
Rotary Screen Separator	100 gal/min	\$50,000	0.6	1.4	EPA AgSTAR (2020), USDA NRCS EQIP (2021), Vendor Data
CHP Unit	500 Nm ³ /hr	\$1,200,000	0.65	1.5	EPA (2021), Vendor Data
RNG Upgrading	500 Nm ³ /hr	\$1,500,000	0.6	1.6	EPA (2021), IEA (2020), Vendor Data

It is important to note that while capital costs are calculated dynamically using the parameters outlined, accurate estimation of total system costs also requires characterization of OPEX for each equipment component. These operational costs include all recurring expenses associated with equipment operation, such as labor, materials, energy consumption, fuels, and water use. Many of these operational parameters are intended to be characterized through the RuFaS biophysical module; however, not all are currently embedded in the model. For operational factors not yet

captured by RuFaS, the model requires user-provided inputs based on available data or site-specific knowledge. By combining these user-defined operational inputs with outputs from the RuFaS biophysical module and the standardized capital costing functions, the model provides a robust, scalable framework for assessing both capital and ongoing operational investments across a range of manure management configurations.

Investments in Animal Comfort

Investments in animal comfort, such as advanced barn ventilation systems, improved milking stalls, and enhanced cooling infrastructure, are fully user-defined within the model. Due to the substantial variability in design, farm size, regional climate, and management practices, these upgrades cannot be captured through a single generalized costing function. Instead, the model provides flexibility for users to characterize these investments using their own data, including capital expenditures, operational costs, financing methods, and expected improvements in herd performance (e.g., milk production gains) and reductions in operational expenses (such as veterinary and health-related costs). This approach enables site-specific analysis of animal comfort strategies when data is available, allowing these improvements to be evaluated alongside manure management system upgrades within an integrated techno-economic framework.

Feed Storage System Improvements

Bunker Silo

TBD

[Others here]

Flexible Economic Framework Equations

The flexible economic framework used in RuFaS includes a suite of equations designed to evaluate the profitability and financial feasibility of management strategies and capital investments. Central to this framework are calculations for ROI, which measure the profitability of a scenario based on changes in cost and revenue, and the DCFROR, which accounts for the time value of money to assess long-term financial outcomes. These methods provide both snapshot and longitudinal views of economic performance. Together, these equations provide a comprehensive, adaptable structure for modeling a wide range of agricultural and energy-related economic scenarios. The following sections detail the specific equations used in each analysis.

Return on Investment Calculation

The return on investment (ROI) calculation is used when a user wants to evaluate the financial merit of a specific change within a partial budget framework. This calculation is designed to estimate how much additional revenue is generated per dollar of additional cost incurred. Users typically apply the ROI tool when assessing the value of adopting a new practice or making any change that alters revenue and cost structures, without including capital investments. ROI is

calculated using Equation 6. While the equation is simple, the insight it provides—expressed as a percentage return—is critical for comparing options and prioritizing changes based on their potential profitability.

$$ROI = \frac{\text{Increase in Revenue} - \text{Increase in Costs}}{\text{Increase in Costs}} \times 100\% \quad (6)$$

Discounted Cashflow Rate of Return (DCFROR) Analysis

The DCFROR analysis is used when a user needs to evaluate the long-term financial viability of a capital investment that spans multiple years. This analysis accounts for the time value of money by calculating the net present value (NPV) of each year within a projected cash flow. By setting the NPV equal to zero, the tool identifies key financial metrics such as the IRR, payback period, or MPSP. These outputs provide essential insight into whether and when an investment becomes profitable, helping guide decisions that extend beyond a single budget year. Table 7 provides a list of all variables used to perform a DCFROR analysis and the equations thereafter are used for DCFROR calculations.

Table 7. Table of user-defined inputs used in the DCFROR (Discounted Cash Flow Rate of Return) analysis.

Parameter Name	Variable	Symbol (for Equations)
cashflow_inputs.target_irr	Target Internal Rate of Return	IRR
cashflow_inputs.target_payback_period	Target Payback period	years
cashflow_inputs.target_product.[item_level_variable]	The <i>item_level_variable</i> for which the minimum product selling price is determined (e.g., milk_class_2)	[<i>item_level_variable</i>]
cashflow_inputs.target_MBSP	Target Minimum Product Selling Price	\$/unit product
cashflow_inputs.[depreciation_i]	Depreciation rate per year	D_r
cashflow_inputs.tax_rate	Tax Rate	T_r
cashflow_inputs.loan_term	Loan Term	L_t
cashflow_inputs.loan_int	Loan Interest Rate	IR_L
cashflow_inputs.const_term	Construction Term	Co_t
cashflow_inputs.const_int	Construction Loan Interest Rate	IR_{Co}
cashflow_inputs.[const_rate_i]	Construction Rate per Year	Co_y
cashflow_inputs.equity	Equity	E

Table 8. Table of model-calculated parameters used in the DCFROR (Discounted Cash Flow Rate of Return) analysis

Variable	Symbol (for Equations)
Financed Amount	$F = 1 - E$
Total Capital Cost	CC_{tot}

Variable	Symbol (for Equations)
Yearly Capital Cost	CC_y
Loan Payment	L_{pay}
Loan Interest Payment	L_{inPay}
Loan Principal	L_{prin}
Revenue	R
Operation Costs	C_{op}
Depreciation Amount	D_{tot}
Net Revenue	R_{net}
Losses Forward	$Loss$
Taxable Income	$I_{taxable}$
Income Tax	T_{income}
Annual Cash Income	A_{income}
Discount Factor	DF
Annual Present Value	A_{PV}
Total Capital Investment + Interest	C_{c+i}
Net Present Value	NPV
Year	y

The DCFROR analysis is structured as a step-by-step process designed to evaluate the long-term financial viability of a capital investment. Each step builds upon the last, enabling the calculation of key financial metrics. The following sections describe the order of operations and the rationale for each equation used in the analysis. An example DCFROR analysis is presented in the **Error! Reference source not found.** section. All assumptions and results are provided in Appendix B (Excel document).

Step 1: Define the Timeline

The analysis begins by establishing the project timeline. If the investment includes a construction period, those years are represented as negative values. For example, a three-year construction phase would include years -2, -1, and 0. The operational phase follows, covering the full duration of the loan term (e.g., years 1 through 30). This approach ensures that pre-revenue capital costs are included in the financial evaluation. Equation 7 is used to establish the range of years used in the analysis, from the start of construction through the final year of the loan. Equation 8 provides an example of a timeline which includes a 3-year construction and at 30-year loan term. These calculations are used to populate the 'Year' column in Appendix B's cash flow table.

$$y = 1 - Co_T: L_t \quad (7)$$

$$y = 1 - 3:30 = -2:30 \quad (8)$$

Step 2: Compute Discount Factors

To account for the time value of money, a discount factor is calculated for each year of the project timeline. This factor adjusts future cash flows to their present value, allowing financial comparisons across time. Equation 9 is used for discount factor calculation. These calculations are used to populate the 'DiscountFactor' column in Appendix B's cash flow table.

$$DF(y) = \frac{1}{(1 + IRR)^y} \quad (9)$$

Step 3: Determine Capital Expenditures During Construction

Capital expenditures are then distributed across the construction years according to a predefined construction rate. This step captures the timing and scale of capital outflows prior to the operational phase. Equation 10 determines the construction time period and Equation 11 calculates annual capital spent. These calculations are used to populate the 'Capital' column in Appendix B's cash flow table.

$$y = 1 - Co_T:0 \quad (10)$$

$$CC_y(y) = Co_y(y) * CC_{tot} * E \quad (11)$$

For each year of capital spending, the portion financed through debt is calculated, Equation 12. This becomes the loan principal and forms the basis for interest and repayment calculations. These calculations are used to populate the 'LoanPrinciple' column in Appendix B's cash flow table.

$$L_{Prin}(y) = Co_y(y) * CC_{tot} * F \quad (12)$$

If interest accrues on the loan during construction, it is calculated using the construction interest rate and the annual loan principal, Equation 13. This cost must be included in the present value of the investment. These calculations are used to populate the 'InterestPayment' column in Appendix B's cash flow table.

$$L_{InPay}(y) = L_{Prin}(y) * IR_{Co} \quad (13)$$

The capital costs and interest payments from the construction period are converted to present value terms using the discount factors, Equation 14. This value is later subtracted from the present value of operational cash flows to determine NPV. These calculations are used to populate the 'NPVCapitalPlusInterest' column in Appendix B's cash flow table.

$$C_{c+i}(y) = [CC_y(y) + L_{InPay}(y)] * DF(y) \quad (14)$$

Step 4: Calculate Annual Loan Payments and Breakdown of Annual Loan Payments

During the operational period (Equation 15), a fixed annual loan payment is computed based on the financed capital, loan interest rate, and loan term (Equation 16). This payment remains constant each year, though its composition (principal vs. interest) changes over time. These calculations are used to populate in the 'LoanPayment' column in Appendix B's cash flow table.

$$y = 1:L_t \quad (15)$$

$$L_{Pay} = \frac{CC_{tot} * IR_L * F}{1 - (1 + IR_L)^{-L_t}} \quad (16)$$

The loan payment is broken into interest and principal components. Early in the loan term, more of the payment goes toward interest; later, more goes toward principal. Equation 17 calculates the loan interest payments and Equation 18 calculates the annual loan principle. These calculations are used to populate the 'InterestPayment' and 'LoanPrinciple' columns, respectively, in Appendix B's cash flow table.

$$L_{InPay}(y) = L_{Prin}(y - 1) * IR_L \quad (17)$$

$$L_{Prin}(y) = L_{Prin}(y - 1) - L_{Pay} + L_{InPay}(y) \quad (18)$$

Step 5: Calculate Depreciation

Annual depreciation is calculated for the capital cost using a predefined schedule using the IRS MACRS method, Equation 19 [5]. This reduces taxable income and is factored into annual cash flow calculations. The MACRS values are shown in the 'DepreciationRate' column in Appendix B's cash flow table. The capital depreciation values populate the 'CapitalDepreciation' column in Appendix B's cash flow table.

$$D_{tot}(y) = D_r(y) * CC_{tot} \quad (19)$$

Step 6: Determine Net Revenue and Taxable Income

Net revenue is computed by subtracting operational costs, loan payments, and depreciation from annual revenue, Equation 20. If net revenue is negative, the value is carried forward as a tax loss, Equation 21. Taxable income is then adjusted accordingly using Equation 22. These calculations are used to populate the 'NetRevenue', 'LossesForward', and 'TaxableIncome' columns, respectively, in Appendix B's cash flow table.

$$R_{net}(y) = R(y) - C_{op}(y) - L_{InPay}(y) - D_{tot}(y) \quad (20)$$

$$Loss(y) = I_{taxable}(y - 1), \text{ if } I_{taxable}(y - 1) < 0 \\ \text{else } 0 \quad (21)$$

$$I_{taxable}(y) = R_{net}(y) + Loss(y) \quad (22)$$

Step 7: Calculate Taxes and Annual Cash Income

Income tax is calculated from taxable income, Equation 23. Annual cash income is then derived by subtracting the tax and loan payment from annual revenue and operation costs, Equation 24. These calculations are used to populate the 'IncomeTax' and 'CashIncome' columns, respectively, in Appendix B's cash flow table.

$$T_{income}(y) = I_{taxable}(y) * T_r \quad (23)$$

$$A_{income}(y) = R(y) - C_{op}(y) - L_{Pay} - T_{income}(y) \quad (24)$$

Step 8: Convert Annual Income to Present Value

Each year's cash income is discounted using the previously calculated discount factor to obtain its present value, Equation 25. These calculations are used to populate the 'PresentValue' column in Appendix B's cash flow table.

$$A_{PV}(y) = A_{income}(y) * DF(y) \quad (25)$$

Step 9: Calculate Net Present Value

Finally, the net present value of the investment is determined by summing all discounted operational cash flows and subtracting the present value of capital costs and construction interest using Equation 26. The total annual net present value is shown in the 'NPVTotal' column in Appendix B's cash flow table. The net present value of tax payment, revenue, operation costs, and loan payments are shown in columns 'NPVTax', 'NPVRevenue', 'NPVOperationCosts', and 'NPVLoanPayment', respectively, in Appendix B's cash flow table.

$$NPV = \sum_{y=1}^{L_t} A_{PV}(y) - \sum_{y=1-CO_t}^0 C_{c+i}(y) \quad (26)$$

Goal Seek Optimization

The DCFROR analysis tool relies on a goal-seek optimization routine to calculate one of three key financial metrics: IRR, payback time, or MPSP. After generating a NPV function by discounting the projected annual cash flows of an investment, users can explore what value of one of these variables would result in an NPV of zero. This process helps identify the conditions under which a capital investment breaks even, making it a valuable tool for evaluating the profitability and feasibility of long-term financial decisions.

Because the NPV equation is non-linear and depends on time-sensitive cash flows, it cannot be solved directly for all variables simultaneously. Users must provide values for any two of the three parameters—IRR, payback period, or MPSP—and the tool will iteratively solve for the third by adjusting its value until NPV converges to zero. For example, with an IRR and desired payback period, the tool will calculate the MPSP required to meet these targets. Alternatively, it can solve for IRR given a set MPSP and payback period, or determine the payback period based on an assumed MPSP and IRR. Users are encouraged to input target values for IRR, payback period, and MPSP, and specify the item-level variable defining the product for which MPSP is calculated. If all three are provided, the tool will perform three goal-seek operations, solving for each parameter with the other two fixed. If only two values are given, it will solve for the third. If only one is provided, the analysis cannot run.

This optimization process can be expressed generically as:

$$\text{Goal Seek: } NPV(IRR, \text{Payback Time}, MPSP) = 0 \quad (27)$$

In practice, the goal seek algorithm evaluates the NPV function over a range of potential values for the unknown variable and adjusts its guess iteratively. For instance, when solving for MPSP, the tool increases or decreases the price across iterations, checking whether the resulting NPV moves closer to zero, and continues adjusting until the difference is negligible. Note that payback time is often constrained to annual intervals, which can limit the granularity of the result. More precise estimates (e.g., monthly payback periods) are technically possible but require more complex implementation and time-resolution of the cash flow data.

Ultimately, this goal seek function allows users to isolate the critical financial thresholds necessary for project feasibility and supports transparent, assumption-driven capital investment planning.

A flow diagram of the DCFROR process including the goal seek optimization is shown in Figure 4.

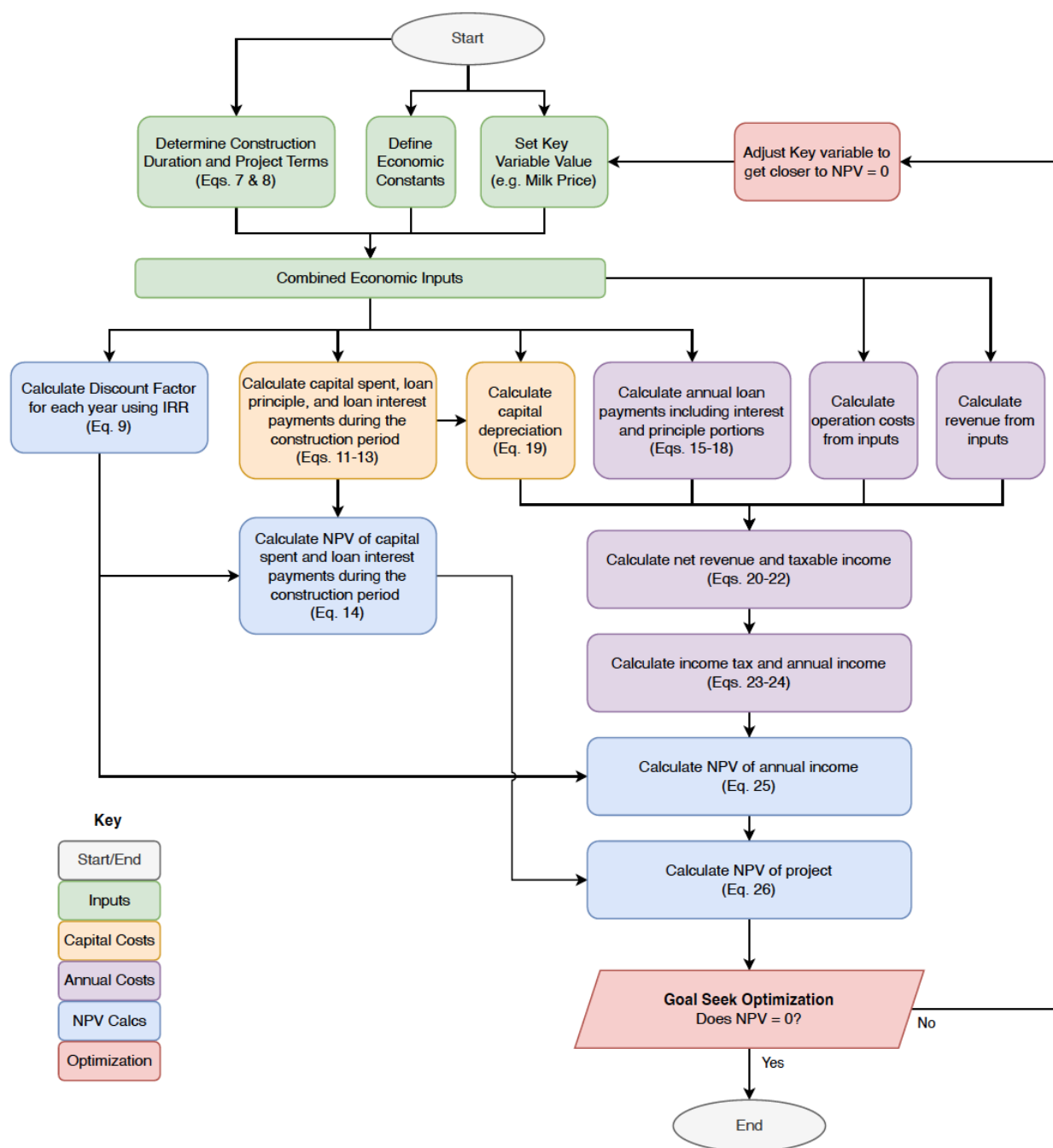


Figure 4. Flow chart of discounted cash flow rate of return analysis calculations

RuFaS Economic Database and Default Values

Data Retrieval Hierarchy

To provide the most accurate and relevant results, the RuFaS Economics Module follows a structured hierarchy for data retrieval. The goal is to use the highest resolution data available for each economic variable. First, the module checks for user-supplied input values. If the user has entered a specific value for a variable, that input takes precedence. If no user input is found, the system searches for historical, spatiotemporal data corresponding to the user's selected location—typically the county in which the analysis is performed—and for the relevant simulation years. If neither user inputs nor location-specific historical data are available, the module defaults to using national average values for the same variable and time period.

This hierarchical approach ensures that user customization is prioritized, while also enabling high-resolution, location-specific data use when available. National average data serves as a fallback, ensuring no gaps in the simulation. Most economic variables in the database are supported by at least one historical value, often sourced from the United States Department of Agriculture (USDA), covering items such as crop prices, product prices, and rates for various farm services.

Economic Data Resolution

Table 9 includes all economic variables used in the analysis, along with their corresponding default values and links to the CSV files containing historical data and that data's source. The table also specifies whether each variable is supported by spatiotemporal data, limited to low-resolution sources, or represents a data gap that requires additional input. While many variables are backed by detailed historical datasets—particularly those related to crop prices, milk prices, and labor rates—others, such as feed supplements, vitamins, and minerals for cattle, are based on single-point estimates or literature values with limited resolution. For variables identified as data gaps, where neither public datasets nor internal expertise are sufficient, RuFaS relies on user input to fill in these values.

All CSV files and the associated data dictionary describing the CSV files can be found in the “_RuFaS Input Files” on the [Sustainability Science RuFaS GitHub repository](#).

Table 9. Data sources for economic variables in the economic database

Variable Name	CSV Filename	Data Category	Reference
commodity_prices.[cow-class_breed_sex]			
commodity_prices.calves_all	commodity_prices.calves_all.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.cows_all	commodity_prices.cows_all.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.cows_milk	commodity_prices.cows_milk.dollar_per_animal.csv	Spatiotemporal	[6]
commodity_prices.cows_ge_500	commodity_prices.cows_ge_500.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.cows_steers_and_heifers_ge_500	commodity_prices.cows_steers_and_heifers_ge_500.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.cow_dairy_fresh	commodity_prices.cow_dairy_fresh.dollar_per_animal.csv	Historical U.S. Averages	[7]
commodity_prices.cow_dairy_bred_t3	commodity_prices.cow_dairy_bred_t3.dollar_per_animal.csv	Historical U.S. Averages	[7]
commodity_prices.cow_dairy_heifer_bred_t3	commodity_prices.cow_dairy_heifer_bred_t3.dollar_per_animal.csv	Historical U.S. Averages	[7]
commodity_prices.cow_dairy_heifer_open	commodity_prices.cow_dairy_heifer_open.dollar_per_animal.csv	Historical U.S. Averages	[7]
commodity_prices.calf_bull_1	commodity_prices.calf_bull_1.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.calf_bull_2	commodity_prices.calf_bull_2.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.calf_heifer_1	commodity_prices.calf_heifer_1.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.calf_heifer_2	commodity_prices.calf_heifer_2.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.steer_holstein_300	commodity_prices.steer_holstein_300.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.steer_holstein_500	commodity_prices.steer_holstein_500.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.steer_holstein_700	commodity_prices.steer_holstein_700.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.[milk_class]			
commodity_prices.milk_all	commodity_prices.milk_all.dollar_per_liter.csv	Spatiotemporal	[6]
commodity_prices.milk_class_2	commodity_prices.milk_class_2.dollar_per_liter.csv	Historical U.S. Averages	[7]
commodity_prices.milk_class_3	commodity_prices.milk_class_3.dollar_per_liter.csv	Historical U.S. Averages	[7]
commodity_prices.milk_class_4	commodity_prices.milk_class_4.dollar_per_liter.csv	Historical U.S. Averages	[7]
commodity_prices.milk_nonfat_dry	commodity_prices.milk_nonfat_dry.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.milk_retail	commodity_prices.milk_retail.dollar_per_liter.csv	Historical U.S. Averages	[8]
commodity_prices.whey_dry	commodity_prices.whey_dry.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.[semen_type]			
commodity_prices.semen_conventional	commodity_prices.semen_conventional.dollar_per_straw.csv	Limited	
commodity_prices.semen_sexed	commodity_prices.semen_sexed.dollar_per_straw.csv	Limited	
commodity_prices.semen_beef	commodity_prices.semen_beef.dollar_per_straw.csv	Limited	
commodity_prices.[bedding_material]			



Variable Name	CSV Filename	Data Category	Reference
commodity_prices.bedding_sand	commodity_prices.bedding_sand.dollar_per_kilogram.csv	Limited	
commodity_prices.bedding_straw	commodity_prices.bedding_straw.dollar_per_kilogram.csv	Limited	
commodity_prices.bedding_sawdust	commodity_prices.bedding_sawdust.dollar_per_kilogram.csv	Limited	
commodity_prices.bedding_manure_solids	commodity_prices.bedding_manure_solids.dollar_per_kilogram.csv	Limited	
commodity_prices.bedding_compost_bedded_pack	commodity_prices.bedding_compost_bedded_pack.dollar_per_kilogram.csv	Limited	
commodity_prices.[feed_form]			
commodity_prices.alfalfa_hay	commodity_prices.alfalfa_hay.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.alfalfa_silage	commodity_prices.alfalfa_silage.dollar_per_kilogram.csv	Scaled Spatiotemporal	[9]
commodity_prices.almond_hulls	commodity_prices.almond_hulls.dollar_per_kilogram.csv	Historical U.S. Averages	[10]
commodity_prices.barley_silage	commodity_prices.barley_silage.dollar_per_kilogram.csv	Scaled Spatiotemporal	[6,11]
commodity_prices.cotton_seed_hulls	commodity_prices.cotton_seed_hulls.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.cotton_seed_meal	commodity_prices.cotton_seed_meal.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.cotton_seed_whole	commodity_prices.cotton_seed_whole.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.corn_grain	commodity_prices.corn_grain.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.corn_silage	commodity_prices.corn_silage.dollar_per_kilogram.csv	Scaled Spatiotemporal	[6,12–14]
commodity_prices.distiller_grains_dried_10pct	commodity_prices.distiller_grains_dried_10pct.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.distiller_grains_modified_wet_50pct	commodity_prices.distiller_grains_modified_wet_50pct.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.distiller_grains_wet_65pct	commodity_prices.distiller_grains_wet_65pct.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.hay_excluding_alfalfa	commodity_prices.hay_excluding_alfalfa.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.hay_all	commodity_prices.hay_all.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.rye_grain	commodity_prices.rye_grain.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.soybean_grain	commodity_prices.soybean_grain.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.soybean_hulls	commodity_prices.soybean_hulls.dollar_per_kilogram.csv	Historical U.S. Averages	[7]
commodity_prices.soybean_meal	commodity_prices.soybean_meal.dollar_per_kilogram.csv	Historical U.S. Market Price	[15]
commodity_prices.sundan_silage	commodity_prices.sundan_silage.dollar_per_kilogram.csv	Limited	[16]
commodity_prices.winter_wheat_grain	commodity_prices.winter_wheat_grain.dollar_per_kilogram.csv	Spatiotemporal	[6]
commodity_prices.calcium_phosphate_di	commodity_prices.calcium_phosphate_di.dollar_per_kilogram.csv	Limited	[17]
commodity_prices.calf_starter_18cp	commodity_prices.calf_starter_18cp.dollar_per_kilogram.csv	Limited	[18]
commodity_prices.limestone	commodity_prices.limestone.dollar_per_kilogram.csv	Limited	[19]
commodity_prices.[crop_seeds]			
commodity_prices.barley_seed	commodity_prices.barley_seed.dollar_per_square_meter.csv	Spatiotemporal	[20]
commodity_prices.corn_seed	commodity_prices.corn_seed.dollar_per_square_meter.csv	Spatiotemporal	[20]
commodity_prices.cotton_seed	commodity_prices.cotton_seed.dollar_per_square_meter.csv	Spatiotemporal	[20]
commodity_prices.oat_seed	commodity_prices.oat_seed.dollar_per_square_meter.csv	Spatiotemporal	[20]
commodity_prices.peanut_seed	commodity_prices.peanut_seed.dollar_per_square_meter.csv	Spatiotemporal	[20]

Variable Name	CSV Filename	Data Category	Reference
commodity_prices.rice_seed	commodity_prices.rice_seed.dollar_per_square_meter.csv	Spatiotemporal	[20]
commodity_prices.sorghum_seed	commodity_prices.sorghum_seed.dollar_per_square_meter.csv	Spatiotemporal	[20]
commodity_prices.soybean_seed	commodity_prices.soybean_seed.dollar_per_square_meter.csv	Spatiotemporal	[20]
commodity_prices.wheat_seed	commodity_prices.wheat_seed.dollar_per_square_meter.csv	Spatiotemporal	[20]
commodity_prices.[net_fertilizer_nutrient]			
commodity_prices.net_fertilizer_nitrogen	commodity_prices.net_fertilizer_nitrogen.dollar_per_kilogram.csv	Historical U.S. Averages	[21,22]
commodity_prices.net_fertilizer_phosphorus	commodity_prices.net_fertilizer_phosphorus.dollar_per_kilogram.csv	Historical U.S. Averages	[21,23]
commodity_prices.net_fertilizer_potassium	commodity_prices.net_fertilizer_potassium.dollar_per_kilogram.csv	Historical U.S. Averages	[21,24]
commodity_prices.net_fertilizer_ammonium_nitrate	commodity_prices.net_fertilizer_ammonium_nitrate.dollar_per_kilogram.csv	Historical U.S. Averages	[21,22]
commodity_prices.net_fertilizer_anhydrous_ammonia	commodity_prices.net_fertilizer_anhydrous_ammonia.dollar_per_kilogram.csv	Historical U.S. Averages	[21,22]
commodity_prices.net_fertilizer_diammonium_phosphate	commodity_prices.net_fertilizer_diammonium_phosphate.dollar_per_kilogram.csv	Historical U.S. Averages	[21,23]
commodity_prices.net_fertilizer_nitrogen_solutions_30pct	commodity_prices.net_fertilizer_nitrogen_solutions_30pct.dollar_per_kilogram.csv	Historical U.S. Averages	[21,22]
commodity_prices.net_fertilizer_potassium_chloride	commodity_prices.net_fertilizer_potassium_chloride.dollar_per_kilogram.csv	Historical U.S. Averages	[21,24]
commodity_prices.net_fertilizer_sulfate_of_ammonium	commodity_prices.net_fertilizer_sulfate_of_ammonium.dollar_per_kilogram.csv	Historical U.S. Averages	[21,22]
commodity_prices.net_fertilizer_super_phosphate_20pct	commodity_prices.net_fertilizer_super_phosphate_20pct.dollar_per_kilogram.csv	Historical U.S. Averages	[21,23]
commodity_prices.net_fertilizer_super_phosphate_44to46pct	commodity_prices.net_fertilizer_super_phosphate_44to46pct.dollar_per_kilogram.csv	Historical U.S. Averages	[21,23]
commodity_prices.net_fertilizer_urea	commodity_prices.net_fertilizer_urea.dollar_per_kilogram.csv	Historical U.S. Averages	[21,22]
commodity_prices.[water_type]			
commodity_prices.water_irrigation	commodity_prices.water_irrigation.dollar_per_cubic_meter.csv	Spatiotemporal	[6]
commodity_prices.water_municipal	commodity_prices.water_municipal.dollar_per_cubic_meter.csv	Spatiotemporal	[25]
commodity_prices.[liquid_fuel_type]			
commodity_prices.diesel	commodity_prices.diesel.dollar_per_liter.csv	Spatiotemporal	[26]
commodity_prices.gasoline	commodity_prices.gasoline.dollar_per_liter.csv	Spatiotemporal	[26]
commodity_prices.[elec_sector]			
commodity_prices.elec_commercial	commodity_prices.elec_commercial.dollar_per_kwh.csv	Spatiotemporal	[27]
commodity_prices.elec_industrial	commodity_prices.elec_industrial.dollar_per_kwh.csv	Spatiotemporal	[27]
commodity_prices.elec_residential	commodity_prices.elec_residential.dollar_per_kwh.csv	Spatiotemporal	[27]
commodity_prices.[natgas_sector]			
commodity_prices.natgas_commercial	commodity_prices.natgas_commercial.dollar_per_megajoule.csv	Spatiotemporal	[28]
commodity_prices.natgas_industrial	commodity_prices.natgas_industrial.dollar_per_megajoule.csv	Spatiotemporal	[28]
commodity_prices.natgas_residential	commodity_prices.natgas_residential.dollar_per_megajoule.csv	Spatiotemporal	[28]

Variable Name	CSV Filename	Data Category	Reference
commodity_prices.[propane_tier]			
commodity_prices.propane_residential	commodity_prices.propane_residential.dollar_per_liter.csv	Spatiotemporal	[29]
commodity_prices.propane_wholesale	commodity_prices.propane_wholesale.dollar_per_liter.csv	Spatiotemporal	[29]
farm_services.[tractor_size]			
farm_services.tractor_small	farm_services.tractor_small.dollar_per_hour.csv		
farm_services.tractor_medium	farm_services.tractor_medium.dollar_per_hour.csv		
farm_services.tractor_large	farm_services.tractor_large.dollar_per_hour.csv		
capital_costs.[tractor_size]			
capital_costs.tractor_small	capital_costs.tractor_small.dollar_per_tractor.csv		
capital_costs.tractor_medium	capital_costs.tractor_medium.dollar_per_tractor.csv		
capital_costs.tractor_large	capital_costs.tractor_large.dollar_per_tractor.csv		
farm_services.labor_hours	farm_services.labor_hours.dollar_per_hour.csv	Spatiotemporal	[6]
farm_services.land_ha_rent	farm_services.land_ha_rent.dollar_per_hectare.csv	Spatiotemporal	[6]
capital_costs.land_ha	capital_costs.land_ha.dollar_per_hectare.csv	Spatiotemporal	[6]
commodity_prices.manure_sales			
commodity_prices.carbon_credits_enteric	commodity_prices.carbon_credits_enteric.dollar_per_tonne.csv		
commodity_prices.carbon_credits_soil	commodity_prices.carbon_credits_soil.dollar_per_tonne.csv		
commodity_prices.digester_rng			
commodity_prices.digester_elec			
commodity_prices.digester_fertilizer			
commodity_prices.digester_bedding			
commodity_prices.digester_carbon_credits			
commodity_prices.rng_upgrading_consumables			

Results Formatting

The economic module produces a variety of outputs to support user interpretation and comparison of scenarios. For DCFROR analyses, RuFaS provides a full cash flow plot across the project life and a detailed table that includes total operational costs, revenue sources, discount factors, net income by year, and NPV by year. When multiple simulations are performed, users also receive a comparison table showing either the delta between cash flow tables or selected columns from individual analyses. This helps users quickly identify which economic factors most influence profitability between scenarios.

For partial budget analyses, the system generates a ROI summary. This includes a stacked bar chart illustrating changes in costs and revenues from a baseline scenario, enabling users to visually assess which variables drive the ROI. In all cases, underlying raw data tables are provided, offering transparency into the numeric values that inform each plot and metric.

Example DCFROR Analysis

This example scenario evaluates the economic feasibility of adding on-site grain storage to a mid-sized dairy farm as a strategy to reduce annual feed purchasing costs. The analysis considers the capital investment required to install two 10,000-bushel steel grain bins along with augers and electrical systems, totaling \$100,000 in upfront costs. It is assumed that 80% of the capital cost is financed, reducing the initial outlay and introducing annual debt service payments into the cash flow analysis. Operating expenses—including labor, electricity, fuel, and maintenance—are estimated at \$5,500 annually. The scenario assumes the farm can purchase feed in bulk at harvest prices, resulting in savings of \$40 per ton compared to purchasing throughout the year, as well as reducing feed losses by 2% annually through improved storage and reduced spoilage. The investment is evaluated using the MACRS with a 7-year depreciation schedule and a target IRR of 10%. Based on these assumptions, we calculate the breakeven feed purchase price necessary for the investment to be financially justified.

In this example, we find that the maximum feed purchase price or minimum crop selling price is greater than the market feed purchase prices (\$672/ton vs \$280/ton), Figure 6. Therefore, it would not be financially profitable to install the feed storage equipment on the farm. Figure 5 provides a breakdown of the annual cashflows associated with this analysis. Figure 6 provides a breakdown of Minimum Product Selling Price into economic components. Figure 7 provides a breakdown of net crop emissions compared to conventional feed production emissions. Figure 8 provides a Sankey diagram breakdown of Minimum Product Selling Price into its economic components.

An Excel file containing all inputs, assumptions, and the resulting DCFROR cashflow table is supplied as Appendix B with this document. Appendix files can also be found on the [Sustainability Science RuFaS GitHub repository](#)

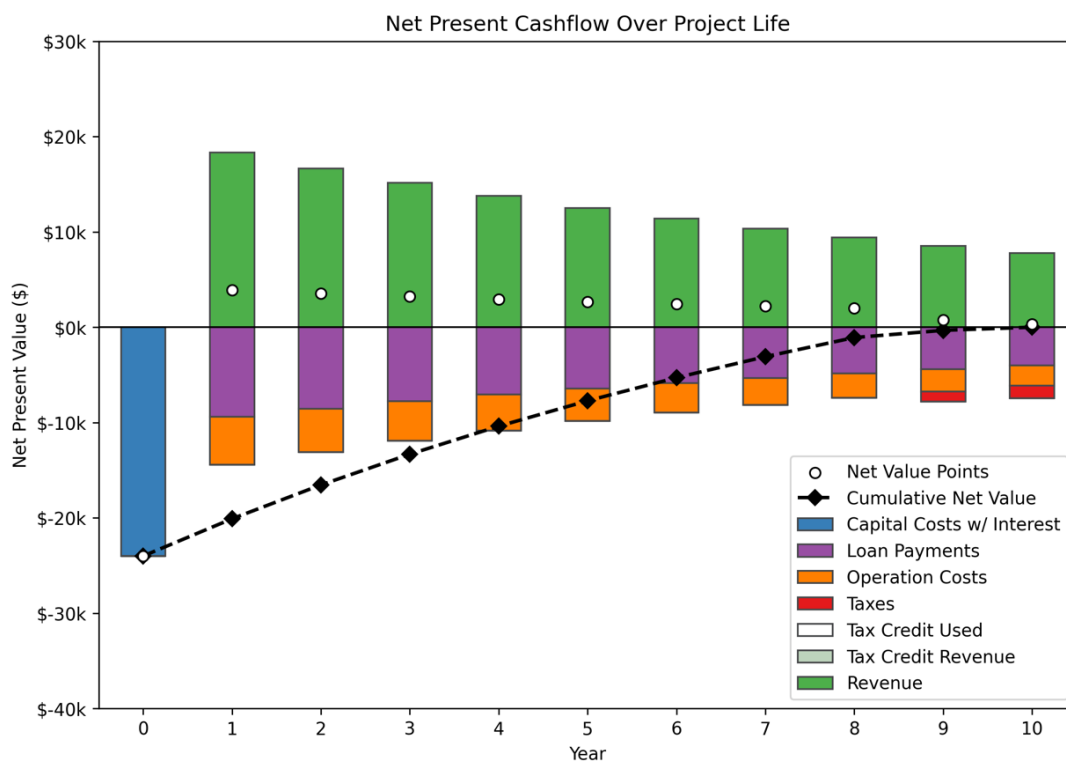


Figure 5. Cashflow for the example DCFROR analysis.

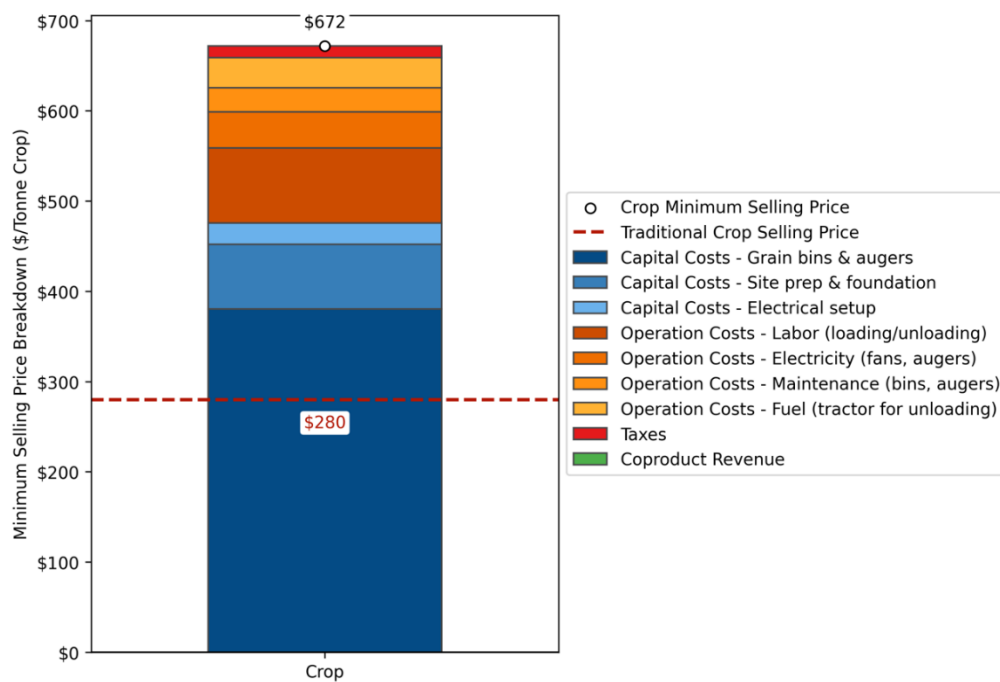


Figure 6. Breakdown of Minimum Product Selling Price into economic components.

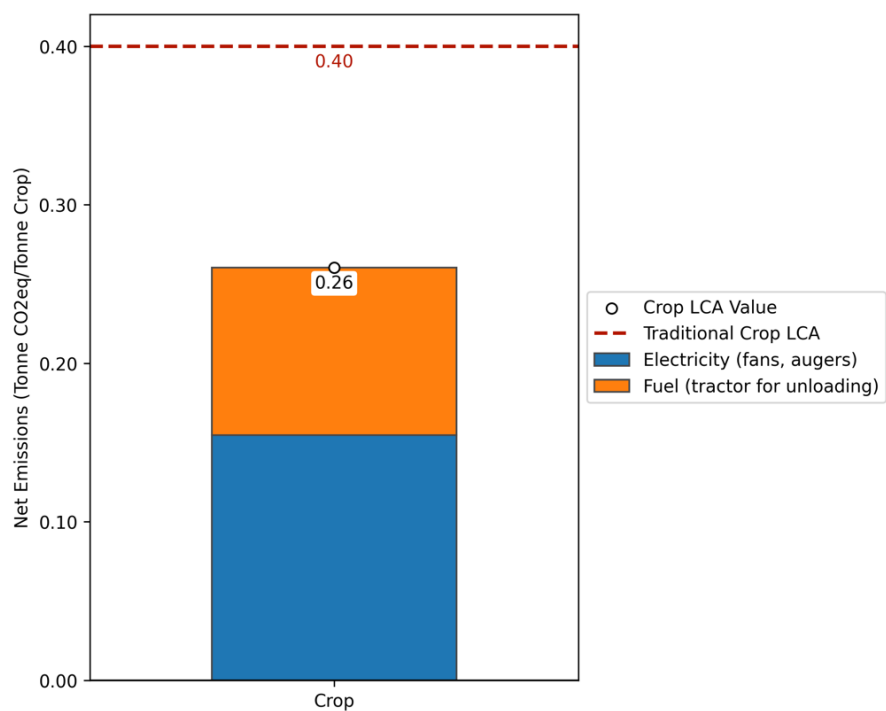


Figure 7. Breakdown of net crop emissions compared to conventional feed production emissions.

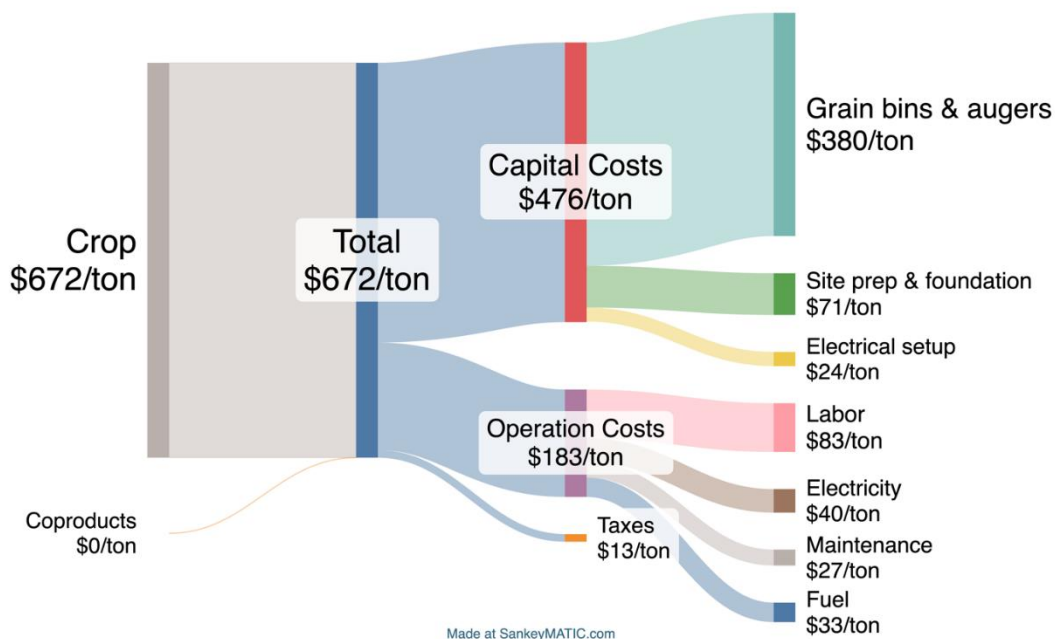


Figure 8. Sankey diagram breakdown of Minimum Product Selling Price into economic component.

Example ROI Analysis

This example assesses the economic return of implementing a higher-cost, more nutritionally rich feed ration for a 200-cow dairy farm operating in a high milk price environment. The farm transitions from a standard feed at \$0.12 per pound to a nutrient-dense option at \$0.15 per pound. With each cow consuming 50 pounds of feed daily, the additional feed cost totals \$109,500 annually. The improved nutrition is projected to increase milk output by 5 pounds per cow per day. Assuming a milk price of \$0.40 per pound—reflecting premiums for organic, grass-fed, or value-added markets—the farm sees an annual revenue increase of \$146,000. This results in a net profit gain of \$36,500 per year and a ROI of 33.3%. The positive return on investment highlights the value of feed optimization when aligned with higher-margin market opportunities and reinforces the importance of integrating both production and pricing dynamics in decision-making. All the assumptions associated with this analysis are shown in Table 10 and the results are shown in Figure 9.

An Excel file containing all inputs, assumptions, and the ROI calculations is supplied as Appendix C with this document. Appendix files can also be found on the [Sustainability Science RuFaS GitHub repository](#).

Table 10. *Economic assumptions associated with the example ROI analysis.*

Scenario A			Scenario B			Return on Investment		
Variable	Value	Units	Variable	Value	Units	Variable	Value	Units
Farm Size	200	cows	Farm Size	200	cows	Increased Costs	\$109,500	\$/yr
Feed Rate	50	lbs/day	Feed Rate	50	lbs/day	Increased Revenue	\$146,000	\$/yr
Standard Feed	\$0.12	\$/lb	Nutrient-dense Feed	\$0.15	\$/lb	ROI	33.3%	
Milk Produced	70	lbs/day	Milk Produced	75	lbs/day			
Milk Price	\$0.40	\$/lb	Milk Price	\$0.40	\$/lb			
Annual Feed Cost	\$438,000	\$/yr	Annual Feed Cost	\$547,500	\$/yr			
Annual Milk Revenue	\$2,044,000	\$/yr	Annual Milk Revenue	\$2,190,000	\$/yr			

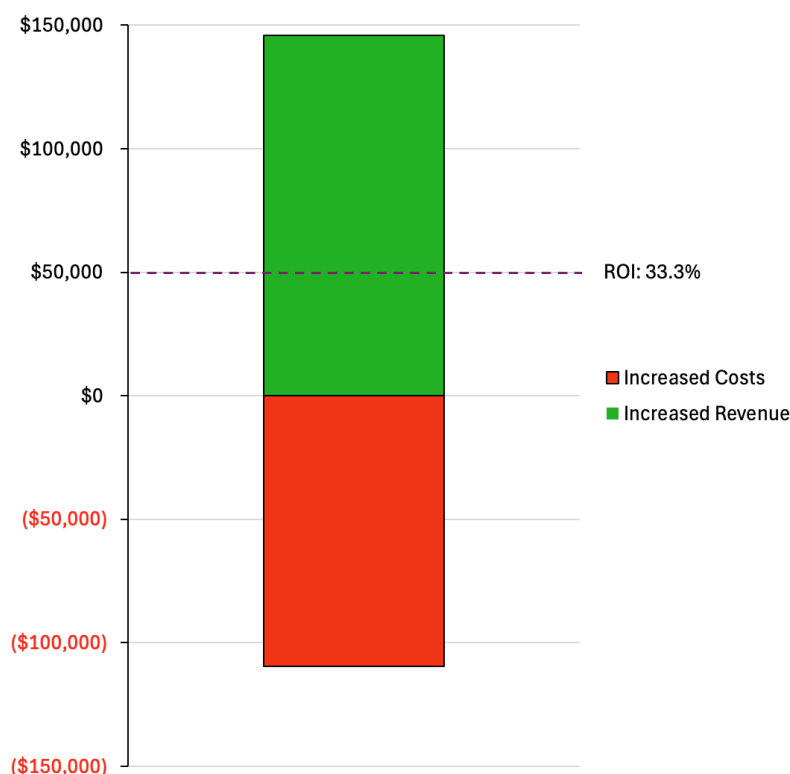


Figure 9. Stacked bar chart showing the results of the example ROI analysis.

Conclusion

This report establishes a comprehensive and transparent framework for evaluating the economic performance of advanced manure management systems. Through the integration of dynamic capital cost functions, user-defined operational parameters, and a flexible discounted cash flow analysis structure, the model enables scenario-based evaluation of farm-scale investments across a wide range of technologies and configurations. By coupling literature-derived digester costing functions with scalable equations for key system components—including solids separators, biogas upgrading systems, and CHP units—the methodology ensures consistency while maintaining flexibility for adaptation to site-specific conditions.

Operational costs are captured with equal rigor, leveraging both user-defined inputs and anticipated outputs from the RuFaS biophysical module to provide an accurate representation of recurring expenses such as labor, energy, water, and material consumption. This modular approach allows the framework to reflect the true cost dynamics of complex manure management systems, and empowers users to assess not only baseline system performance but also the

economic impacts of technology upgrades, process optimizations, and management interventions.

Importantly, the model is designed for extensibility. It accommodates a range of user-defined practices, such as investments in more advanced manure management systems and enhanced feeding strategies, enabling a holistic assessment of on-farm interventions that influence both productivity and sustainability outcomes. By quantifying trade-offs between capital investment, operational expenses, and revenue opportunities—such as energy sales, nutrient recovery, and participation in emerging carbon markets—the framework equips producers, advisors, and policymakers with actionable insights to guide investment decisions.

Together, these capabilities position the model as a powerful decision-support tool, capable of supporting industry stakeholders in pursuing economically viable pathways toward improved manure management, reduced environmental impact, and enhanced farm resilience under a range of future scenarios.

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Appendix A: Variable Sub-Categories

Variable Class	Variable Name	Description	Units
[cow-class_breed_sex]	calves_all	Cattle prices were gathered from the United States Department of Agriculture's (USDA's) National Agricultural Statistics Service Quickstats web portal	\$/kg
	cows_all	Cattle prices were gathered from the United States Department of Agriculture's (USDA's) National Agricultural Statistics Service Quickstats web portal	\$/kg
	cows_milk	Cattle prices were gathered from the United States Department of Agriculture's (USDA's) National Agricultural Statistics Service Quickstats web portal	\$/kg
	cows_ge_500	Cattle prices were gathered from the United States Department of Agriculture's (USDA's) National Agricultural Statistics Service Quickstats web portal	\$/kg
	cows_steers_and_heifers_ge_500	Cattle prices were gathered from the United States Department of Agriculture's (USDA's) National Agricultural Statistics Service Quickstats web portal	\$/kg
	cow_dairy_fresh	Price for Fresh Dairy Cow Replacements (recently gave birth and lactating) with weight 1,000-1,600 lbs from USDA's National Dairy Comprehensive Report	\$/animal
	cow_dairy_bred_t3	Price for T3 Bred Dairy Cow (mature and pregnant) with weight 1,000-1,600 lbs from USDA's National Dairy Comprehensive Report	\$/animal
	cow_dairy_heifer_bred_t3	Price for T3 Bred Dairy Heifer (pregnant and never calved before) with weight 800-1,400 lbs from USDA's National Dairy Comprehensive Report	\$/animal
	cow_dairy_heifer_open	Price for Open Dairy Heifer (not yet been bred or is not currently pregnant) with weight 600-900 lbs from USDA's National Dairy Comprehensive Report	\$/animal
	calf_bull_1	Price for No.1 Bull Calves (0-14 days) with weight 90-130 lbs from USDA's National Dairy Comprehensive Report	\$/kg
	calf_bull_2	Price for No.2 Bull Calves (0-14 days) with weight 85-120 lbs from USDA's National Dairy Comprehensive Report	\$/kg
	calf_heifer_1	Price for No.1 Heifer Calves (0-14 days) with weight 80-130 lbs from USDA's National Dairy Comprehensive Report	\$/kg
	calf_heifer_2	Price for No.2 Heifer Calves (0-14 days) with weight 85-120 lbs from USDA's National Dairy Comprehensive Report	\$/kg
	steer_holstein_300	Price for Feeder Holstein Steers (300-500 lbs) from USDA's National Dairy Comprehensive Report	\$/kg
	steer_holstein_500	Price for Feeder Holstein Steers (500-700 lbs) from USDA's National Dairy Comprehensive Report	\$/kg
	steer_holstein_700	Price for Feeder Holstein Steers (700-1,000 lbs) from USDA's National Dairy Comprehensive Report	\$/kg
[feed_form]	alfalfa_hay	Alfalfa hay price received by farmers gathered from the United States Department of Agriculture's (USDA's) National Agricultural Statistics Service Quickstats web portal	\$/kg

Variable Class	Variable Name	Description	Units
	alfalfa_silage	Alfalfa silage prices were scaled off corn silage prices based on “Costs of Forage Production” by Rotz and Harrigan	\$/kg
	almond_hulls	Historical prices for almond hulls were gathered from the USDA’s Agricultural Marketing Service	\$/kg
	barley_silage	Barley silage prices were scaled off corn silage prices based on “Silage Cost of Production” by the Beef Cattle Research Council	\$/kg
	cotton_seed_hulls	Average price paid for cotton seed hulls as a feed by farmers from USDA’s National Dairy Comprehensive Report	\$/kg
	cotton_seed_meal	Average price paid for cotton seed meal as a feed by farmers from USDA’s National Dairy Comprehensive Report	\$/kg
	cotton_seed_whole	Average price paid for whole cotton seeds as a feed by farmers from USDA’s National Dairy Comprehensive Report	\$/kg
	corn_grain	Average price received for corn grain by farmers from USDA’s NASS	\$/kg
	corn_silage		
	distiller_grains_dried_10pct	Average price paid for Dried Distillers Grains (10% moisture) as a feed by farmers from USDA’s National Dairy Comprehensive Report	\$/kg
	distiller_grains_modified_wet_50pct	Average price paid for Modified Wet Distillers Grains (50-55% moisture) as a feed by farmers from USDA’s National Dairy Comprehensive Report	\$/kg
	distiller_grains_wet_65pct	Average price paid for Wet Distillers Grains (65-70% moisture) as a feed by farmers from USDA’s National Dairy Comprehensive Report	\$/kg
	hay_excluding_alfalfa	Hay price received by farmers excluding alfalfa hay was gathered from the United States Department of Agriculture’s (USDA’s) National Agricultural Statistics Service Quickstats web portal	\$/kg
	hay_all	Hay price received by farmers was gathered from the United States Department of Agriculture’s (USDA’s) National Agricultural Statistics Service Quickstats web portal	\$/kg
	rye_grain	Rye grain price received by farmers gathered from the United States Department of Agriculture’s (USDA’s) National Agricultural Statistics Service Quickstats web portal	\$/kg
	soybean_grain	Soybean grain price received by farmers gathered from the United States Department of Agriculture’s (USDA’s) National Agricultural Statistics Service Quickstats web portal	\$/kg
	soybean_hulls	Average price paid for soybean hulls as a feed by farmers from USDA’s National Dairy Comprehensive Report	\$/kg
	soybean_meal	Daily soybean meal commodity prices were gathered from Business Insider at their historical market prices	\$/kg
	sundan_silage	Sudan silage prices were estimated from “2009 Sample Costs to Produce Sudan Silage for San Joaquin Valley – South” by Wright et al. produced by the University of California Cooperative Extension	\$/kg
	winter_wheat_grain	Winter wheat grain price received by farmers gathered from the United States Department of Agriculture’s (USDA’s) National Agricultural Statistics Service Quickstats web portal	\$/kg

Variable Class	Variable Name	Description	Units
[milk_class]	milk_all	Average price received for all milk by farmers from USDA's NASS	\$/liter
	milk_class_2	Average price received for class 2 milk (used for soft dairy products like yogurt, ice cream) by farmers from USDA's National Dairy Comprehensive Report	\$/liter
	milk_class_3	Average price received for class 3 milk (used for cheese production) by farmers from USDA's National Dairy Comprehensive Report	\$/liter
	milk_class_4	Average price received for class 4 milk (used for butter and nonfat dry milk) by farmers from USDA's National Dairy Comprehensive Report	\$/liter
	milk_nonfat_dry	Average price received for nonfat dry milk by farmers from USDA's National Dairy Comprehensive Report	\$/kg
	milk_retail	Average retail price paid for whole milk from Federal Reserve Bank of St. Louis FRED Economic Data	\$/liter
	whey_dry	Average price received for dry whey by farmers from USDA's National Dairy Comprehensive Report	\$/kg
[crop_seeds]	barley_seed	Average regional seed price per area of planted land as provided by the USDA's Economic Research Service	\$/m ²
	corn_seed	Average regional seed price per area of planted land as provided by the USDA's Economic Research Service	\$/m ²
	cotton_seed	Average regional seed price per area of planted land as provided by the USDA's Economic Research Service	\$/m ²
	oat_seed	Average regional seed price per area of planted land as provided by the USDA's Economic Research Service	\$/m ²
	peanut_seed	Average regional seed price per area of planted land as provided by the USDA's Economic Research Service	\$/m ²
	rice_seed	Average regional seed price per area of planted land as provided by the USDA's Economic Research Service	\$/m ²
	sorghum_seed	Average regional seed price per area of planted land as provided by the USDA's Economic Research Service	\$/m ²
	soybean_seed	Average regional seed price per area of planted land as provided by the USDA's Economic Research Service	\$/m ²
	wheat_seed	Average regional seed price per area of planted land as provided by the USDA's Economic Research Service	\$/m ²